

Overview

SnorkeJ

Rainier - University Math Guidelines

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How to Get Started

1. Navigate to the [Expert Platform](#)
2. Once signed in, go to **My Projects**
3. Search for **"Rainier,"** and locate the following projects:
 - a. **EK_Rainier_Production_BTAD / Rainier Submission**
 - b. **EK_Rainier_Production_BTAD / Rainier Review**
4. Click the blue "Start" button to begin!

Rainier Submission

Project: EK_Rainier_Production_BTAD

Start

Rainier Review

Project: EK_Rainier_Production_BTAD

Start

Submission

Rainier Submission Guidelines

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Overview

Welcome to the **Rainier Math Project**! You'll be helping us develop a **dataset of new, model-stumping math problems** across thirteen domains of mathematics.

Task Description

Each annotation task consists of creating an original, text-only (LaTeX-formatted) math problem which is run through **frontier AI models to determine difficulty** and contains the following components:

- **Problem:** A clear, single, self-contained text-only problem **written in LaTeX** that contains a math question that can be used to evaluate the reasoning capabilities of models. The problems must have an open solution path but a closed answer space.
 - **Note that you should not be asking multiple independent questions in your problem and linking them together. A problem is ONE question!**
- **Answer:** The final correct answer that answers the problem
- **Domain Classification and Explanation:** Your problem must fit into one of the [13 available domains and corresponding sub-domains](#) and include a **justification** as to **why** it is the most appropriate selection
- **Step-by-Step Solution:** A step-by-step worked solution (also written in LaTeX) to the problem written by experts. The final answer should appear in a LaTeX boxed environment in the final step

- **Problem Type, Answer Type, and Concepts:** Select a description from a list of possible types that best describes the problem and answer.

 **DO NOT USE LLMs TO GENERATE ANY PART OF YOUR SUBMISSION. EVIDENCE OF THIS WILL RESULT IN AN OUTRIGHT REJECTION AND REMOVAL FROM THE PROJECT!**

Available Domains

There are **13 Domains** to choose from on this project. Each problem should be assigned to the Domain and Sub-Domain that **best describe** the problem.

These domains should reflect the central subject matter of the **problem**, not just the notation used, the final answer format, or techniques that may appear in one possible solution.

See the [Domain Description](#) tab for further details:

1. **Algebra, Functions, and Trigonometry**
 2. **Euclidean, Coordinate, and Transformational Geometry**
 3. **Calculus**
 4. **Linear Algebra**
 5. **Abstract Algebra**
 6. **Number Theory**
 7. **Discrete Mathematics and Combinatorics**
 8. **Probability and Statistics**
 9. **Differential Equations and Dynamical Systems**
 10. **Analysis**
 11. **Topology and Geometry**
 12. **Optimization and Numerical Mathematics**
 13. **Logic, Set Theory, and Foundations**
-

Quality Requirements

To be considered a high-quality submission eligible for Acceptance, the prompt must meet **all** of the following:

- **Original:** All components must be authored by human experts (LLM generated content is not permitted), and all components within each sample must be original
- **Appropriate Math Domains:** Use mathematical concepts at in the [allowed domains](#)
- **Challenging:** Stumps at least one frontier AI model majority of the time.

- **Requires reasoning:** The problem must require step-by-step reasoning to answer, rather than reciting memorised facts, theorems, etc. Solutions should be **at least 3 steps**
- **Well-formed:** All components must be grammatically or syntactically correct, and without spelling or [LaTeX formatting errors](#)
- **Unique answer exists:** The problem must only have a single unique solution, but it may have multiple equivalent valid forms
 - **“Does not exist” or DNE is not a valid answer on this project**
- **Clear answer format:** The prompt should explicitly state the specified format of the answer. If the answer is meant to be parametric or expressed in terms of other variables, that should be clearly stated.
- **Reliably verifiable:** Answers to the problems and Solution Steps **MUST** be verifiable
- **Self-contained:** Aside from the expected math knowledge, problems must contain all the necessary context required to answer the problem. Tables, reference material, etc. that are required for answering must be included in the problem
- **No shortcutting:** The problem must be free of hints or shortcuts that make it “trivial” to answer
- **No guessable answers:** The problem must not be easily guessable (e.g. yes/no or multiple choice)
- **No proofs:** Prompts that ask “show that”, “prove”, “verify”, etc. are not allowed.
- **Problem makes sense:** The problem itself must make sense in the context of the selected domain
- **Hand-Solveable:** Questions must be solvable by hand without the use of calculators or computers. For example, **the following are not allowed:**
 - Working with numbers larger than 10^{15} or smaller than 10^{-10}
 - Require computations that are impossible or unreasonable to do by hand (e.g. evaluate $\sin(5)$ or e^{-3})
 - Implementing iterative algorithms or writing code to obtain the answer
- **Problems should not rely on overly obscure/niche topics, for example:**
 - Gromoll-Meyer spheres
 - Rolfsen tables
 - Alexander polynomials
- **Note:** Word puzzles are **not allowed**, even if solvable with set theory.

Answer Requirements

To be considered a high-quality submission eligible for Acceptance, the **answer must be correct**.

Additionally, **the answer cannot use any notation that hasn't been introduced in the prompt**. Otherwise, this violates the self-contained requirement.

Solution Requirements

To be considered a high-quality submission eligible for Acceptance, the **step-by-step solution** must meet the the following requirements:

- **Correctness:** All components must be mathematically valid and arrive at the final answer.
 - **Valid Individual Steps:** Verify each solution step independently of the answer and recompute key displayed equalities and numeric value yourself. A correct final answer does not certify the steps!
 - **All Claims are Supported:** For every nontrivial claim (especially after 'clearly', 'note that', 'it follows') confirm the argument is present. Check that cited theorems actually apply, that induction is proven (not pattern-matched), and that case splits cover all cases including boundary/sign/degenerate ones.
 - **Accurate and Consistent Formatting:** All components must be grammatically or syntactically correct, and without spelling or LaTeX formatting errors. All expressions, variables, equations, etc. are formatted consistently across all steps of the solution.
 - [See Appendix LaTeX formatting](#) for more details
 - **Project Conventions:**
 - Each step must start with "Step #"
 - The last step must end with "Final Answer: $\boxed{\dots}$ "
 - The answer in the "Answer Box" must match the final answer in the last step.
-

Pro Tips for High-Quality Samples

- Ensure you are aligned with the above formatting requirements.
 - **Double-check** your questions are distinct from one another and are not from existing datasets/competitions.
 - Use good grammar, check your spelling, etc.
 - Ensure your **Solution** does not include steps **within a step**, or any **trivial micro-steps**.
 - These should be broken up as much as possible, and should only cover the most **significant or substantial steps** made to reach the correct solution. You do not have to include steps that were not important in reaching the solution or points that aren't necessary in breaking down these substantial steps.
 - Defer to your best judgement here as to what should/should not be included. You can refer to the **good example steps** in the [Examples](#) tab as a reference for how these should be written.
-

LaTeX Help

Having issues with LaTeX formatting? Here are a ton of helpful resources you can utilize:

- [Overleaf](#): offers **command completion** as well as a side by side source and **preview** - a great resource to reference whilst working through formatting issues

- [Codecogs](#): for those **unfamiliar with** LaTeX syntax, this is an equation editor where you can click on the symbols you wish to write, and see the LaTeX code in a window, and the equation formatted in another. Please note that you will need to copy that LaTeX code and enclose it within $\dots$ or $\$ \$ \dots \$ \$$ in your prompt or solution!
- [Detextify](#): for those **just beginning to learn** LaTeX syntax, here you can hand draw symbols which are then returned with the LaTeX code for those symbols
- [Dainiak](#): a **quick** LaTeX style **checker**
- [TeX Studio](#): integrated writing environment **for creating LaTeX documents** with features including syntax-highlighting, integrated viewer, reference checking, etc.

We also have four helpful youtube videos introducing LaTeX that could be helpful as well:

- [What is LaTeX?](#)
- [Your first LaTeX document](#)
- [Basic mathematics in LaTeX](#)
- [Intermediate mathematics in LaTeX](#)

Detailed Submission Instructions

To Start

- In a **Rainier Submission**:
 - You will start by testing your math problem against two **frontier-level “hard” models** that are intentionally difficult to stump. These models may take a few minutes to generate a response.
 - **Click the blue arrow** to run your question through the model and **ensure BOTH models consistently get the wrong answer**.
 - Enter your math problem in LaTeX in the text box that states: *“Enter your prompt here.”*

Rainier Submission

UID: 74ddea68-11f9-4717-a4a7-d4fab532001c

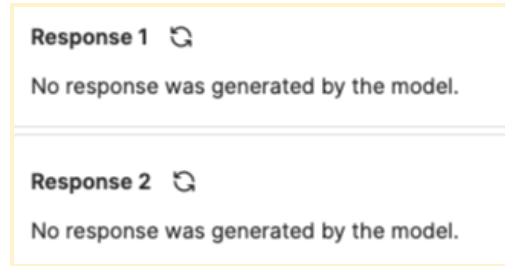
Note: These are for your reference when writing your problem and are not the official difficulty check. It may be possible for a model to get the wrong answer here, once, but get the problem correct most of the time. When in doubt, make sure your problem stumps both of these models consistently. If you get “No Response” please try re-running by clicking the arrow icon and let the Snorkel team know if you encounter consistent issues.

Please write your problem here to test it against AI models (These are thinking models and may take a few minutes to run).

Enter your prompt here

→

⚠ PLEASE NOTE: “No Response was generated by the model” is not equivalent to the model getting an incorrect answer. If you encounter this, please re-run the models. If you consistently get this error, please reach out to the Snorkel Team via Slack.



Response 1 ↻

No response was generated by the model.

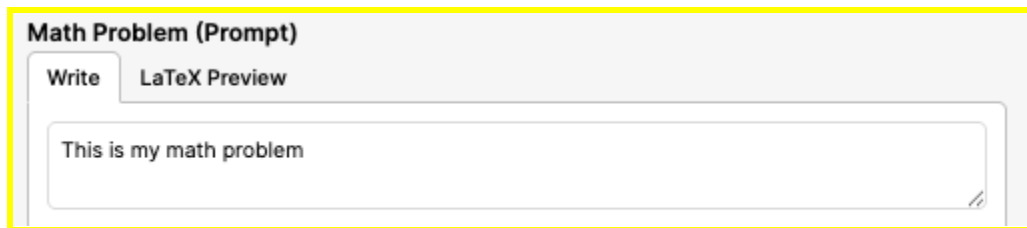
Response 2 ↻

No response was generated by the model.

Submitting the Question

Once you're confident you've created a problem that **consistently stumps the provided models**, you'll complete the rest of the task on the right side of the screen.

- Use the copy and paste your prompt into the text box under the “**Math Problem (Prompt)**” heading on the right-hand side of the UI.



Math Problem (Prompt)

Write LaTeX Preview

This is my math problem

- Select the appropriate [domain and sub-domain](#) for the question:
 - Algebra, Functions, and Trigonometry
 - Euclidean, Coordinate, and Transformational Geometry
 - Calculus
 - Linear Algebra
 - Abstract Algebra
 - Number Theory
 - Discrete Mathematics and Combinatorics
 - Probability and Statistics
 - Differential Equations and Dynamical Systems
 - Analysis
 - Topology and Geometry
 - Optimization and Numerical Mathematics
 - Logic, Set Theory, and Foundations

 **Note on Domain Quotas:** Each of these domains and sub-domains have quotas, and will stop accepting new submissions once that quota has been reached. Submissions with the wrong domain selected may be rejected outright if the correct category's quota is full.

- Fill out the **Domain Explanation** field by providing a brief and clear explanation of why this domain and subdomain are the best choice for your math problem and why it's more appropriate than the next best domain and subdomain. Justifications should follow the below structure:

This problem involves <description of the key mathematical elements of the problem>,

which are part of <domain and sub domain>.

The problem also involves <description of the additional mathematical elements of the problem>,

which are part of <other domain>.

However, <explanation of why they're not core to the problem or less advanced>.

For Example: *This problem involves a finite symmetric difference evolution whose exact solution is encoded by polynomial generating functions, which is part of Discrete Mathematics and Combinatorics and Generating functions. The problem also involves finite quotient rings and cyclic update operators, which are part of Abstract Algebra. However, those algebraic tools only support the generating function calculation that determines the exact residue profile, so they are not the main subject of the problem.*

Worked Solution

- Provide a step-by-step solution in which each step makes a substantial advance toward the final answer; avoid breaking the solution into many trivial micro-steps.

The final step of the problem must use LaTeX box format to denote the answer and end with **"Final Answer: $\boxed{\text{<answer goes here>}}$ "**

- Ensure that each step of your solution can be verified by the Reviewer **independently of the answer** - recompute key displayed equalities and numeric values yourself prior to submitting. **A correct final answer does not certify the steps!**
 - For every nontrivial claim (especially after 'clearly', 'note that', 'it follows') **confirm your argument is present**. Check that cited theorems actually apply, that induction is proven (not pattern-matched), and that case splits cover all cases including boundary/sign/degenerate ones prior to submitting!

- Provide **all necessary reasoning steps to advance the solution** in the “**Worked Solution**” field.
 - Recall that this can include introducing a crucial substitution, applying an important theorem, making a necessary case split, or combining earlier results to reach the correct solution.
 - There should be a minimum of at least **3** steps.
 - Each **step** must make a significant advancement toward reaching the final answer rather than simply breaking down the solution into small/trivial pieces.
 - Do not include steps that were not inherently important or points that are not necessary to understanding how the solution was reached.
 - Ensure each Step is just that, **one step**. Do not include **steps within a step, substeps, micro-trivial steps, or small/trivial content** that isn't an **absolutely important** or necessary step in reaching the final answer.

 **DO NOT USE LLMs TO GENERATE ANY PART OF YOUR WORKED SOLUTION STEPS. EVIDENCE OF THIS WILL RESULT IN AN OUTRIGHT REJECTION!**

- You **must** enumerate each worked solution step using the following format: *Step 1, Step 2, Step 3, etc*
- All worked solution steps must be **explicitly numbered in sequence** (*Step 1, Step 2, ...*)
- If you are referencing a **previous step** (i.e., discussing something from Step 2 that is being recalled or referenced in Step 4, you should explicitly mention that as “Step 2”)

Step-by-Step Solution

Provide a **step-by-step** solution in which each step makes a **substantial advance** toward the final answer; avoid breaking the solution into many trivial micro-steps.

Each step must start with “**Step #:**” (e.g. Step 1; Step 2; etc.)

The final step of the problem must **use LaTeX box format** to denote the answer and end with “**Final Answer:** $\boxed{\text{<answer goes here>}}$ ”

1

Write

LaTeX Preview

Step #:

2

Write

LaTeX Preview

Step #:

3

Write

LaTeX Preview

Step #:

Answer Section

- Add in the final answer to the problem (and do **NOT** use $\boxed{\text{ }}$ in this field!)
- Once done, run the In-App Evaluation button to validate your submission

- Please disregard the “(optional)” portion of the header; validating your submission is **mandatory for this project!**

Answer
The final answer to the problem. (Do **not** use `\boxed` for this, but do wrap your answer in `$...$`)

Write LaTeX Preview

Input field text

In-App Evaluation (optional)
Please run the following helper tool to validate your submission.

- See the [Accepted Examples](#) tab for good examples to reference!

Problem, Answer, and Concepts Section

- Lastly, you will select the most appropriate problem and answer classification. Select the options that best describe your problem based on the [Answer Types](#) and [Problem Types](#) descriptions above!
- You will also be asked to list out the **concepts, techniques, or knowledge** required to solve your problem. This must include **at least one and no more than five**. If more than five apply, you'll be asked to list the **five most important**. Lists should follow the below structure:

For **example**, these concepts could be listed for a calculus problem:

- *Critical points*
- *First-derivative sign changes*
- *Local maxima/minima*
- *Endpoint comparison*
- *Polynomial differentiation*

Problem Type, Answer Type, and Concepts



Submission Routing

In this section, you will be able to select **where would you like your submission** to go after you submit, with the following options:

- Select **"Quality Review"** when you believe your submission **meets project quality standards**, passes automated checks, and is ready for quality review by a human reviewer
- Or, you can select the optional **"Automated Checks Only"** to check your **in-progress submission** against the automated checks for **difficulty and originality**. This will be an indicator as to whether or not your submission is ready for quality review!

Submission Routing



Automated Difficulty and Quality Feedback

If you do elect to send your submission through the optional **"Automated Checks Only,"** you will be able to see the results of these automated checks while still working on your submission, including:

- Whether your submission is of acceptable difficulty
- The full results of how the models solved your problems
- Whether your problem and solution are correct and complete
- Whether your solution steps are correct and if there are logical gaps or mathematical errors
- Whether your selected domain and justification are appropriate

Automated Difficulty and Quality Feedback



- For example, one of the checks:

Automated Difficulty Feedback (optional)

These are the results from Snorkel's test of your submission against AI models, and if your submission is of acceptable difficulty.

B I U ↻

Automated Difficulty Check Results

Result: ● Your submission is sufficiently difficult

Model 1 Success Rate: 0%

Model 2 Success Rate: 0%

Note: Any edits or eventual revisions to your Problem or Answer could result in your problem's difficulty changing.

WARNING: Your problem may have an incorrect answer. The tested model consistently got the following answer:

0.16

Final Checks

- **Final Checklist:** Confirm the following:
 - ☒ There is **one and only one correct answer**.
 - ☒ You have **checked for spelling and grammar**.
 - ☒ You have **independently verified the accuracy of your solution steps**.
 - ☒ The question is **not plagiarized**.
 - ☒ You **did not use LLMs** to generate any part of your submission.
 - ☒ You have used appropriate **LaTeX formatting**.

Submission

- When all fields are complete, click **Submit**.

Review Process Updates

Review Process Updates: PLEASE READ

For those who have worked on previous Snorkel math projects, the review process has undergone several changes aimed at improving and clarifying this workflow so **all expectations are explicitly clear** and reviews are of the **highest possible quality**. Please read through the following updates:

#1. Double-Down on Solution Steps:

The solution steps are often the largest reason for a task not meeting the Acceptance bar, and being sent back for revision. Moving forward, **Reviewers must do the following**:

- Verify each step independently of the answer and recompute key displayed equalities and numeric value yourself. A **correct final answer does not certify the steps!**
 - For every **nontrivial claim** (especially after 'clearly', 'note that', 'it follows') **confirm the argument is present**. Check that cited theorems actually apply, that induction is proven (not pattern-matched), and that case splits cover all cases including boundary/sign/degenerate ones.
-

#2. Selecting the New “Error Categories”:

To improve the consistency of review feedback, we have implemented **new Error Categories** in your review tasks that will allow you to **select specific labels for errors you identify**.

Reviewers will be required to select these categories and ensure their revision feedback corresponds with what was selected.

- See [Error Category Definitions](#)
 - Reviewers also **MUST** provide specific, actionable suggestions to revise any identified error categories. Feedback like *“make the problem more self-contained”* will **not be tolerated** on this project.
 - Every time a Reviewer leaves feedback, the question should be: **If the Original Author addressed everything I included in the revision notes, would it be acceptable upon re-submission?** If not, **more** needs to be added!
 - See [Good v Bad Feedback](#) below for more details
-

#3. Mandatory Automated Feedback Check

Reviewers will now be required to look over the complete automated feedback checks on a submission **in a review task**. The purpose of this is to provide reviewers more information/context on the submission itself, as well as how our automated checks work to help **improve your understanding** of each aspect of the review workflow. Doing these checks will allow reviewers to be more comfortable working with evals:

- Upon selecting “**Accept**” for the Review Decision, Reviewers will be prompted with answering another question: **Are All Automated Checks Accurate?**
- Reviewers will note whether any of the automated checks are **inaccurate**, including false positives, but indicate why the task can **still be Accepted**
- See more details in the [Step-by-Step System Walkthrough](#) tab!

Review Decision

☒ Accept
 ☐ Needs Revision
 ☐ Reject

Are All Automated Checks Accurate?

Please review the results of the blue "Automated Feedback" box and the "Automated Difficulty and Quality Feedback" section and determine if any feedback is incorrect.

For example, are any issues flagged by the checks that aren't actually wrong?

☐ Yes
 ☐ No

#4. Transparent Reviewer Performance Feedback

In Rainier, Reviewers will be provided a transparent performance feedback system that will allow for continued improvement and closer collaboration with our team to ensure alignment on expectations. This will specifically include **Weekly “Summaries”** to Reviewers on review performance, including: errors that were missed, areas of misalignment, areas to improve on, areas of strength, and a full performance breakdown. We will track overall performance, which will be grouped into the following statuses:

- **Trial Period**
 - Once a Reviewer passes the Reviewer Assessment, they will begin with a “**Trial Period**” of **3-5 days**. This will include a **throttled daily review limit** to allow for a smooth transition into reviewing as you adjust to the guidelines/workflow.
- **Unlimited (No Throttle)**
 - After the Trial Period, once a Reviewer has consistently produced high-quality reviews and demonstrated a clear, comprehensive understanding of the review workflow and expectations, they will be given “**Unlimited**” status. Reviewers will be able to complete the maximum amount of daily reviews, **40**.
- **At Risk (Throttled)**
 - With the Weekly Summaries provided, Reviewers will also be able to see the **% of problems they reviewed that were found to have had errors** compared to their

total review count. If a Reviewer reaches more than **10% of errors** across all reviews (which includes **ANY** errors that slipped through, including minor issues like LaTeX) this will warrant a **violation**, and they will be placed in an “**At Risk**” status.

- This violation will include a **significant decrease of the Reviewer’s daily limit**, as well as an individual coaching message with specific examples of the errors identified to assist the Reviewer in improving. This status will be lifted once the Reviewer demonstrates consistent improvement on the previously identified errors.
- If a Reviewer’s error percentage **continues to increase during “At Risk” status**, and improvement is not shown, this will potentially result in a **removal of review access in Rainier**.

Note: a violation can also be warranted if revision feedback is found to be generic, vague, unhelpful, or misaligned with the expectations outlined [here](#).

Problem and Answer Reference

Problem Types

The “ask” of the problem should broadly fit into one of the buckets defined below. There may be other problem forms, as this list is not meant to be exhaustive:

1. **Exact computation:** A problem asking for an exact numerical value or exact mathematical object derived from the given information
2. **Symbolic derivation:** A problem asking for a closed-form expression, identity, recurrence, transform, or asymptotic form obtained through symbolic reasoning
3. **Numerical approximation:** A problem asking for a numerical answer computed to a specified tolerance, precision, or error bound

Problem Type Bonus Categories

4. **Solve for unknowns:** A problem asking for one or more values or objects that satisfy a given equation, system, or mathematical condition
 5. **Construction under constraints:** A problem asking for the unique mathematical object that satisfies a stated set of structural or quantitative constraints.
 6. **Optimization:** A problem asking for an optimal value, an optimizer, or both under a given objective and constraint set
 7. **Exhaustive enumeration:** A problem asking for the complete set of all valid solutions, cases, or objects satisfying the given criteria
 8. **Canonicalization or normalization:** A problem asking for an object to be rewritten in a specified standard, reduced, or canonical form
 9. **Transformation between representations:** A problem asking for conversion of an object from one mathematically equivalent representation to another
 10. **Parameter identification:** A problem asking for unknown parameters, coefficients, or latent quantities to be inferred from equations, conditions, or observed structure
 11. **Other:** Describe a type beyond those listed above
-

Answer Types

The “type” of answer that is expected for each problem should broadly fit into one of the buckets defined below. There may be other answer types, and this list is not meant to be exhaustive.

Generally, we expect that answers can be categorized into the following types:

1. **Exact scalar:** A single exact numerical quantity, such as an integer, rational number, algebraic number, or closed form constant

2. **Numerical scalar approximation:** A single numerical quantity expected up to a specified tolerance or precision. Any tolerances must be specified in the problem
3. **Exact symbolic expression:** A closed-form symbolic expression representing the correct final answer up to mathematical equivalence

Answer Type Bonus Categories

4. **Polynomial or rational function:** A polynomial, rational function, or closely related algebraic expression returned in symbolic form
5. **Equation or inequality:** An equation, inequality, system, recurrence, congruence, or constraint written explicitly as the answer.
6. **Interval or region description:** An interval, union of intervals, domain, range, feasible region, solution region, or set described by inequalities or symbolic conditions.
7. **Vector:** A finite-dimensional vector returned as coordinates, components, or an equivalent explicit representation
8. **Matrix:** A matrix returned explicitly, typically with entries in exact or numerical form as specified.
9. **Tuple or ordered list:** A finite ordered collection of values, objects, labels, or coordinates where order matters.
10. **Set or multiset of objects:** A complete finite collection of valid outputs, where order is irrelevant and multiplicity may or may not matter
11. **Sequence or series representation:** A sequence, recurrence, finite truncation rule, or series-form representation of the target object
12. **Function or mapping:** A function, transformation, homomorphism, permutation, operator, rule, or mapping specified explicitly by formula, table, images of elements, or action on a basis.
13. **Canonical form:** A mathematical object expressed in a specified standard or normalized form, such as diagonal form, Jordan form, or reduced row echelon form
14. **Other:** Describe a type beyond those listed above

Domain Reference

Domain and Sub-Domain Descriptions

 Please note that Domain and Sub-Domains will continually close as the project progresses - meaning they will be unavailable to submit questions to once the target number for that particular selection is filled. You can stay up to date on the most recent closures by checking the Workflow Resources Canvas and scrolling down to “Closed Topics”

1. Algebra, Functions, and Trigonometry

Problems involving algebraic structure, symbolic manipulation, elementary functions, equations, inequalities, graphs, and trigonometric computation. Avoiding and excluding proof-based algebra, abstract algebraic structures, and problems whose primary concept is calculus, geometry, or numerical approximation.

- **Algebraic expressions and manipulation** - Simplifying, factoring, expanding, rationalizing, and rewriting algebraic expressions.
- **Equations and inequalities** - Solving linear, quadratic, polynomial, rational, radical, exponential, logarithmic, absolute-value, and trigonometric equations or inequalities.
- **Functions and graphs** - Working with domains, ranges, composition, inverse functions, transformations, intercepts, asymptotes, and graph features.
- **Polynomial and rational functions** - Analyzing roots, factors, multiplicities, end behavior, asymptotes, and rational-function structure.
- **Exponential and logarithmic functions** - Solving and interpreting problems involving exponential growth, decay, logarithmic identities, and logarithmic equations.
- **Trigonometry** - Evaluating trigonometric expressions, using identities, solving trigonometric equations, and analyzing trigonometric graphs.
- **Sequences and elementary recurrence relations** - Working with arithmetic, geometric, and simple recursively defined sequences.
- **Systems of equations** - Solving linear and nonlinear systems, including parameter-dependent systems.
- **Elementary mathematical modeling** - Representing quantitative situations using algebraic, exponential, logarithmic, or trigonometric functions.

2. Euclidean, Coordinate, and Transformational Geometry

Problems involving classical geometry, coordinate-based geometry, geometric measurement, and transformations. Avoiding and excluding abstract topology, manifold theory, proof-based synthetic geometry, and problems whose primary concept is calculus or linear algebra.

- **Euclidean geometry** - Computing and classifying properties of angles, triangles, circles, polygons, congruence, similarity, and classical metric relationships.

- **Coordinate and analytic geometry** - Working with distances, slopes, midpoints, lines, circles, conics, and equations of geometric objects in coordinate systems; studying curves, conics, intersections, tangents, normals, and loci using algebraic methods.
- **Solid geometry** - Computing volumes, surface areas, cross-sections, and spatial relationships among three-dimensional objects.
- **Transformational geometry** - Applying translations, rotations, reflections, dilations, symmetries, and compositions of transformations.
- **Trigonometric geometry** - Using right-triangle trigonometry, law of sines, law of cosines, and trigonometric area formulas.
- **Computational geometry** - Solving finite geometric problems involving points, lines, polygons, convex hulls, Voronoi diagrams, and geometric algorithms.

3. Calculus

Problems involving limits, differentiation, integration, infinite series, and calculus-based analysis of functions. Avoiding and excluding proof-based analysis, abstract measure-theoretic questions, and differential equations as the primary object of study.

- **Limits and continuity** - Evaluating limits, one-sided limits, infinite limits, asymptotes, and continuity properties.
- **Differentiation** - Computing derivatives, higher derivatives, implicit derivatives, logarithmic derivatives, and derivatives using standard rules.
- **Applications of derivatives** - Solving problems involving optimization, related rates, monotonicity, concavity, tangent lines, linear approximation, and extrema.
- **Integration** - Computing antiderivatives, definite integrals, improper integrals, and integrals using standard techniques.
- **Applications of integration** - Computing areas, volumes, arc length, average value, work, mass, probability densities, and accumulated quantities.
- **Infinite sequences and series** - Analyzing convergence, power series, Taylor series, radius of convergence, and series approximations.
- **Multivariable calculus** - Working with partial derivatives, gradients, directional derivatives, multiple integrals, constrained extrema, and changes of variables.
- **Vector calculus** - Computing with vector fields, line integrals, surface integrals, divergence, curl, Green's theorem, Stokes' theorem, and the divergence theorem.

4. Linear Algebra

Problems involving vectors, matrices, finite-dimensional vector spaces, linear transformations, and spectral computations. Avoiding and excluding abstract module theory, functional analysis, and proof-based structural classification unless the answer is a directly checkable computation or classification.

- **Vectors and vector spaces** - Working with linear combinations, span, independence, bases, dimension, and coordinates.

- **Matrices and matrix operations** - Performing matrix arithmetic, computing inverses, transposes, ranks, and block matrix operations.
- **Systems of linear equations** - Solving systems using row reduction, Gaussian elimination, consistency checks, and parameterized solution forms.
- **Linear transformations** - Computing kernels, images, matrix representations, rank-nullity quantities, and change-of-basis representations.
- **Determinants** - Computing determinants and using them to test invertibility, volume scaling, and matrix identities.
- **Eigenvalues and eigenvectors** - Computing characteristic polynomials, eigenvalues, eigenspaces, diagonalizations, and spectral data.
- **Inner product spaces** - Computing orthogonality, projections, Gram-Schmidt bases, least-squares solutions, norms, and angles.
- **Matrix decompositions and canonical forms** - Working with LU, QR, SVD, spectral decomposition, Jordan form, and related normal forms.
- **Numerical linear algebra** - Applying computational methods for linear systems, eigenvalue problems, conditioning, stability, and large-scale approximation.
- **Tensor and multilinear algebra** - Computing with tensor products, bilinear forms, multilinear maps, and coordinate representations.

5. Abstract Algebra

Problems involving algebraic structures such as groups, rings, fields, modules, and representations. Avoiding and excluding long proof-based prompts; problems should ask for computations, classifications of concrete structures, explicit examples, or finite algebraic outputs.

- **Group theory** - Working with groups, subgroups, cyclic groups, permutation groups, cosets, quotient groups, homomorphisms, and group actions.
- **Ring theory** - Working with rings, ideals, quotient rings, units, zero divisors, polynomial rings, and ring homomorphisms.
- **Field theory** - Computing with fields, field extensions, finite fields, irreducibility, algebraic elements, and extension degrees.
- **Module theory** - Working with modules over rings, submodules, quotient modules, free modules, and module homomorphisms.
- **Galois theory** - Computing or identifying splitting fields, Galois groups, automorphisms, and solvability-related algebraic data.
- **Representation theory** - Working with group representations, characters, invariant subspaces, and decompositions into irreducibles.
- **Universal algebra and algebraic structures** - Working with semigroups, monoids, lattices, Boolean algebras, and related algebraic systems.
- **Computational algebra** - Computing Gröbner bases, symbolic polynomial reductions, finite algebraic structures, and algorithmic algebra outputs.
- **Commutative algebra** - Primary decomposition, localization, dimension theory, completion, Hilbert-Samuel functions and series, etc.

- **Homological Algebra** - Ext and Tor functors, chain complexes, derived functors, resolutions

6. Number Theory

Problems involving integer structure, divisibility, modular arithmetic, Diophantine constraints, and arithmetic functions. Avoiding and excluding proof-based number theory unless the expected answer is a directly checkable value, classification, example, counterexample, or finite set.

- **Elementary number theory** - Working with divisibility, primes, greatest common divisors, least common multiples, Euclidean algorithm, and prime factorization.
- **Modular arithmetic and congruences** - Solving congruences, computing modular inverses, applying the Chinese remainder theorem, and evaluating modular exponentiation.
- **Diophantine equations** - Finding or classifying integer solutions to linear, quadratic, polynomial, and exponential equations.
- **Multiplicative functions** - Computing Euler's phi function, divisor functions, Möbius function, and related arithmetic functions.
- **Quadratic residues and reciprocity** - Working with Legendre symbols, Jacobi symbols, residue classes, and quadratic congruences.
- **Analytic number theory** - Computing or estimating prime-counting functions, zeta-related quantities, Dirichlet series values, and asymptotic arithmetic behavior.
- **Algebraic number theory** - Working with number fields, rings of integers, ideals, norms, traces, class groups, and units.
- **Computational number theory** - Applying primality tests, factorization algorithms, modular algorithms, and cryptographic arithmetic computations.

7. Discrete Mathematics and Combinatorics

Problems involving finite or countable structures, counting, graphs, relations, and discrete systems. Avoiding and excluding proof-only combinatorial arguments unless the answer is a count, construction, classification, example, counterexample, or finite object.

- **Basic counting principles** - Applying permutations, combinations, inclusion-exclusion, pigeonhole principle, and counting with restrictions.
- **Enumerative combinatorics** - Counting structured objects, computing partitions, and evaluating combinatorial identities.
- **Generating functions** - Using ordinary or exponential generating functions, coefficient extraction, and recurrence-solving methods.
- **Graph theory** - Working with paths, cycles, trees, connectivity, colorings, matchings, flows, planarity, and graph invariants.
- **Extremal combinatorics** - Finding maximum or minimum possible sizes of discrete objects under specified constraints.

- **Probabilistic combinatorics** - Analyzing random discrete structures and probability-based counting problems.
- **Design theory and finite geometry** - Working with block designs, finite incidence structures, Latin squares, and combinatorial configurations.
- **Discrete structures** - Working with relations, equivalence relations, partial orders, and finite automata.
- **Matroid theory** - Computing with independence systems, rank functions, circuits, bases, and representable matroids.

8. Probability and Statistics

Problems involving uncertainty, random variables, distributions, statistical inference, and data-based reasoning. Avoiding and excluding open-ended data interpretation, essay-style statistical critique, and proof-based probability theory unless the final answer is directly verifiable.

- **Probability foundations** - Computing with events, sample spaces, conditional probability, independence, Bayes' rule, and probability axioms.
- **Random variables and distributions** - Working with discrete and continuous distributions, cumulative distribution functions, densities, moments, and transformations.
- **Joint distributions and dependence** - Computing marginals, conditionals, covariance, correlation, independence, and multivariate distribution quantities.
- **Expectation and variance** - Computing expected values, variances, covariances, moment-generating functions, and expectation identities.
- **Stochastic processes** - Working with Markov chains, Poisson processes, martingales, random walks, Brownian motion, and time-indexed random systems.
- **Limit theorems** - Applying laws of large numbers, central limit theorem, and convergence in probability, distribution, or mean.
- **Statistical inference** - Computing estimators, confidence intervals, hypothesis tests, p-values, likelihoods, and sampling distributions.
- **Bayesian statistics** - Computing priors, likelihoods, posteriors, conjugate updates, posterior summaries, and Bayesian decision rules.
- **Regression and statistical modeling** - Working with linear regression, generalized linear models, residuals, model fit, and prediction.
- **Experimental design and causal inference** - Computing or identifying randomization structures, confounding, treatment effects, blocking, and causal estimands.

9. Differential Equations and Dynamical Systems

Problems involving equations of change, time evolution, stability, and qualitative or numerical behavior of dynamic systems. Avoiding and excluding proof-based existence theory unless the answer is a checkable classification, solution, parameter value, or computed quantity.

- **First-order ordinary differential equations** - Solving separable, linear, exact, autonomous, and Bernoulli equations.

- **Higher-order ordinary differential equations** - Solving linear ODEs, constant-coefficient equations, variation of parameters problems, and undetermined-coefficients problems.
- **Systems of differential equations** - Working with linear systems, phase portraits, matrix exponentials, and coupled dynamics.
- **Partial differential equations** - Solving or classifying heat, wave, Laplace, transport, and other equations involving partial derivatives.
- **Boundary value problems** - Solving differential equations with boundary constraints, eigenvalue conditions, and Sturm-Liouville structure.
- **Initial value problems** - Computing explicit solutions, approximate solutions, and solution values from initial data.
- **Dynamical systems** - Analyzing iterated maps, flows, fixed points, periodic orbits, phase space, and long-term behavior.
- **Stability theory** - Classifying stability of equilibria, linearizations, Lyapunov behavior, and sensitivity to perturbation.
- **Bifurcation theory** - Identifying parameter-dependent qualitative changes in equilibria, cycles, and system behavior.
- **Numerical differential equations** - Applying Euler methods, Runge-Kutta methods, finite differences, finite elements, and stability checks for schemes.

10. Analysis

Problems involving rigorous properties of functions, limits, spaces, convergence, integration, and operators. Avoiding and excluding proof-based prompts; problems should focus on classification, computation, examples, counterexamples, or checking hypotheses of named results.

- **Real analysis** - Working with sequences, limits, continuity, differentiability, Riemann integration, uniform convergence, and real-valued function behavior.
- **Metric spaces** - Classifying open and closed sets, compactness, completeness, connectedness, convergence, and continuity in metric spaces.
- **Measure theory** - Working with sigma-algebras, measurable functions, measures, Lebesgue integrals, convergence theorems, and product measures.
- **Functional analysis** - Working with normed spaces, Banach spaces, Hilbert spaces, bounded linear operators, dual spaces, and compact operators.
- **Complex analysis** - Computing with analytic functions, residues, contour integrals, Laurent series, conformal maps, and singularities.
- **Fourier analysis** - Working with Fourier series, Fourier transforms, convolution, frequency-domain representations, and orthogonal expansions.
- **Harmonic analysis** - Analyzing function decompositions, maximal functions, singular integrals, and analysis on groups or spaces.
- **Asymptotic analysis** - Computing Big-O relationships, asymptotic expansions, limiting approximations, perturbation approximations, and extreme-parameter behavior.
- **Calculus of variations** - Computing functionals, Euler-Lagrange equations, extremal functions, and variational optimization quantities.

11. Topology and Geometry

Problems involving abstract spaces, manifolds, curvature, topological invariants, and advanced geometric structures. Avoiding and excluding elementary Euclidean geometry as the primary topic and proof-based topology unless the expected answer is a directly checkable invariant, classification, construction, or example.

- **Point-set topology** - Working with topological spaces, open sets, bases, continuity, compactness, connectedness, and separation axioms.
- **Algebraic topology** - Computing or identifying homotopy groups, fundamental groups, homology, cohomology, covering spaces, and topological invariants.
- **Differential topology** - Working with smooth manifolds, tangent spaces, transversality, embeddings, immersions, and degree theory.
- **Differential geometry** - Computing properties of curves, surfaces, manifolds, curvature, connections, geodesics, and geometric tensors.
- **Riemannian geometry** - Working with Riemannian metrics, geodesic distance, curvature tensors, sectional curvature, and Riemannian volume.
- **Symplectic and contact geometry** - Working with symplectic forms, Hamiltonian systems, contact structures, and canonical transformations.
- **Geometric topology** - Working with manifolds, knots, links, mapping class groups, and low-dimensional topological structures.
- **Convex geometry** - Working with convex sets, polytopes, support functions, separation results, and convex bodies.
- **Algebraic geometry** - Working with varieties, schemes, ideals, coordinate rings, morphisms, divisors, and computational algebraic geometry.

12. Optimization and Numerical Mathematics

Problems involving optimization, approximation, computational mathematics, numerical stability, and algorithmic solution methods. Avoiding and excluding open-ended algorithm design or empirical performance discussion unless the expected answer is a computed value, classification, selected method, or finite output.

- **Linear programming** - Solving problems with linear objectives, linear constraints, feasible regions, simplex method, duality, and sensitivity.
- **Convex optimization** - Working with convex sets, convex functions, KKT conditions, duality, projections, and proximal methods.
- **Nonlinear optimization** - Solving unconstrained and constrained nonlinear optimization problems using gradients, Hessians, Lagrange multipliers, and local extrema.
- **Discrete and integer optimization** - Solving optimization problems over finite, combinatorial, or integer-constrained feasible sets.
- **Dynamic programming and optimal control** - Computing Bellman recurrences, sequential decisions, control constraints, and optimal trajectories.
- **Numerical analysis** - Approximating roots, integrals, derivatives, interpolants, and solutions to continuous mathematical problems.

- **Approximation theory** - Working with polynomial approximation, splines, orthogonal approximations, and best approximation problems.
- **Error analysis and stability** - Computing conditioning, roundoff error, truncation error, convergence rates, and algorithmic stability properties.
- **Numerical optimization** - Applying gradient descent, Newton methods, quasi-Newton methods, stochastic optimization, and convergence diagnostics.

13. Logic, Set Theory, and Foundations

Problems involving formal systems, set-theoretic structure, computability, complexity, and foundational mathematical objects. Excluding informal proof-writing or philosophical discussion; problems should use truth values, finite formal manipulations, classifications, constructions, or explicit examples.

- **Mathematical logic** - Working with propositional logic, predicate logic, formal languages, syntax, semantics, truth tables, and logical equivalence.
- **Proof theory** - Working with formal properties of deductive systems, derivation rules, sequent calculi, consistency, and normalization as mathematical objects.
- **Model theory** - Working with structures, interpretations, satisfaction, definability, elementary equivalence, and theories of mathematical structures.
- **Set theory** - Working with sets, relations, functions, cardinality, ordinals, cardinals, well-orderings, and axiomatic set systems.
- **Computability theory** - Working with Turing machines, decidability, recursively enumerable sets, reductions, and computable functions.
- **Complexity theory** - Working with complexity classes, reductions, completeness, time complexity, space complexity, and computational hardness.
- **Category theory** - Working with categories, functors, natural transformations, limits, colimits, adjunctions, and categorical constructions.
- **Type theory and formal systems** - Working with typed calculi, formal terms, judgments, normalization, and computational interpretations of logic.

Accepted Examples

Accepted Examples

Example Problem #1

Domain: Probability and Statistics

Sub-Domain: Limit Theorems

Problem Type: Parameter Identification

Solution Type: Numerical Scalar Approximation

PROBLEM PROMPT:

1. Suppose $U_1, \dots, U_n$ are independent random variables with density $f_{U_i}(u | \theta) = \theta u^{-(\theta+1)} I(u \geq 1)$, where $\theta > 0$ and $I(u \geq 1)$ denotes the indicator function, equal to 1 when $u \geq 1$ and 0 otherwise.
2. Define $V_i = U_i^{-2}$, for $i = 1, \dots, n$.
3. Let $V_{(1)} \leq \dots \leq V_{(n)}$ be the order statistics of $V_1, \dots, V_n$.
4. Define $R_n = \max_{1 \leq i \leq n} \ln U_i$.
5. Let $\theta = 2$.
6. Let $a > 0$, $b \in \mathbb{R}$ be constants such that $an^b(V_{(n)} - e^{-2R_n} - 2V_{(n-1)} + V_{\lfloor \ln n \rfloor})$ converges in distribution to a non-degenerate random variable with variance 1 as $n \rightarrow \infty$. Here, $\lfloor \ln n \rfloor$ is the flooring of $\ln n$.
What is the value of $a + b$?
Give only a decimal rounded to 4 decimal places.

DOMAIN JUSTIFICATION:

The problem involves convergence in distribution of a normalized sequence of random variables which is within the domain of Probability and Statistics. The problem also involves choosing constants so that the limiting random variable is non-degenerate with variance 1 which is part of Limit Theorems.

STEP-BY-STEP SOLUTION & ANSWER:

Step-by-Step Solution

Step 1: Since U_i has density $f_U(u | \theta) = \theta u^{-(\theta+1)} I(u \geq 1)$, we have, for $u \geq 1$, $\mathbb{P}(U_i > u) = u^{-\theta}$. When $\theta = 2$, this becomes $\mathbb{P}(U_i > u) = u^{-2}$. Define $V_i = U_i^{-2}$. Then, for $0 < v \leq 1$, $\mathbb{P}(V_i \leq v) = \mathbb{P}(U_i^{-2} \leq v) = \mathbb{P}(U_i \geq v^{-1/2}) = (v^{-1/2})^{-2} = v$. Therefore $V_i \sim \text{Unif}(0, 1)$.

Step 2: By definition, $R_n = \max_{1 \leq i \leq n} \ln U_i = \ln \max_{1 \leq i \leq n} U_i$. Hence $e^{-2R_n} = e^{-2 \ln \max_i U_i} = (\max_i U_i)^{-2}$. Since $V_i = U_i^{-2}$ is a decreasing transformation of U_i , the largest U_i corresponds to the smallest V_i . Therefore $e^{-2R_n} = V_{(1)}$. Thus the expression inside the parentheses is $V_{(n)} - e^{-2R_n} - 2V_{(n-1)} + V_{\lfloor \ln n \rfloor} = V_{(n)} - V_{(1)} - 2V_{(n-1)} + V_{\lfloor \ln n \rfloor}$.

Step 3: Since $V_1, \dots, V_n$ are i.i.d. $\text{Unif}(0, 1)$, we have $V_{(n)} \rightarrow 1$, $V_{(n-1)} \rightarrow 1$, and $V_{(1)} \rightarrow 0$ in probability. Therefore $V_{(n)} - V_{(1)} - 2V_{(n-1)} \rightarrow 1 - 0 - 2 = -1$ in probability.

Step 4: Notice that the last term in parentheses has no parentheses around the index, so it is not the order statistic $V_{(\lfloor \ln n \rfloor)}$. It is the original observation with index $\lfloor \ln n \rfloor$. Since each $V_i \sim \text{Unif}(0, 1)$, we have $V_{\lfloor \ln n \rfloor} \sim \text{Unif}(0, 1)$ for every n . Hence, by Slutsky's theorem, $V_{(n)} - V_{(1)} - 2V_{(n-1)} + V_{\lfloor \ln n \rfloor} \Rightarrow -1 + V$, where $V \sim \text{Unif}(0, 1)$.

Step 5: The quantity inside the parentheses already converges to the non-degenerate random variable $V - 1$. Therefore, no power of n is needed. If $b < 0$, then $n^b \rightarrow 0$, and the limit would

become degenerate at 0. If $b > 0$, then $n^b \rightarrow \infty$, and the sequence would not converge to a proper non-degenerate random variable. Hence, we must have $b = 0$.

Step 6: With $b = 0$, the limiting distribution is $a(V - 1)$, where $V \sim \text{Unif}(0, 1)$. Since $\text{Var}(V - 1) = \text{Var}(V) = 1/12$, the condition that the limiting variance equals 1 gives $\text{Var}(a(V - 1)) = a^2 \text{Var}(V - 1) = a^2/12 = 1$. Therefore $a^2 = 12$, and since $a > 0$, $a = \sqrt{12} = 2\sqrt{3}$.

Step 7: Therefore $a + b = 2\sqrt{3} + 0 = 3.4641016\dots$. Rounded to four decimal places, the final answer is 3.4641.

Final Answer: 3.4641

Answer
3.4641

Example Problem #2

Domain: Logic and Foundations

Sub-Domain: Mathematical Logic

Problem Type: Classification by Exhaustive Solution Set

Solution Type: Vector

PROBLEM PROMPT:

Let

$$L = \{E, P, Q\},$$

where E is a binary relation symbol and P, Q are unary relation symbols.

For an L -structure, define the four color-types

$$00, \quad 10, \quad 01, \quad 11$$

according to the truth values of P and Q .

Let $\mathcal{K}$ be the class of finite L -structures satisfying:

1. E is an equivalence relation. 2. There are exactly 12 equivalence classes. 3. Every equivalence class has exactly 10 elements. 4. Each equivalence class has one of the following six color-profiles, written in the order $(00, 10, 01, 11)$:

$$\tau_1 = (10, 0, 0, 0), \quad \tau_2 = (6, 4, 0, 0), \quad \tau_3 = (5, 5, 0, 0), \quad \tau_4 = (4, 4, 2, 0), \quad \tau_5 = (4, 4, 1, 1), \quad \tau_6 = (3, 3, 2, 2).$$

For $M \in \mathcal{K}$, define $m(M) = (m_1, m_2, m_3, m_4, m_5, m_6)$, where m_i is the number of E -classes in M having profile τ_i .

Fix a structure $M_0 \in \mathcal{K}$ with $m(M_0) = (1, 2, 3, 1, 4, 1)$.

For $M, N \in \mathcal{K}$, write $M \equiv_4 N$ if M and N satisfy the same first-order L -sentences of quantifier rank at most 4.

Determine the complete set of vectors $m(N)$ such that $N \in \mathcal{K}$ and $N \equiv_4 M_0$.

DOMAIN JUSTIFICATION:

This problem belongs to Logic and Foundations, specifically Mathematical logic, because the central question is bounded first-order indistinguishability of finite structures. More precisely, the natural topic is finite model theory: the problem asks what first-order sentences of quantifier rank at most 4 can distinguish about finite equivalence-relation structures with unary predicates. Since Model theory is not an available sub-domain option, Mathematical logic is the closest available choice. The problem also involves a finite counting classification of color-profile multiplicities, which is combinatorial, but that counting is secondary to the bounded first-order equivalence analysis.

STEP-BY-STEP SOLUTION & ANSWER:

Step-by-Step Solution

Step 1:

Write

$$\chi_{00}(x) = \neg P(x) \wedge \neg Q(x), \quad \chi_{10}(x) = P(x) \wedge \neg Q(x), \quad \chi_{01}(x) = \neg P(x) \wedge Q(x), \quad \chi_{11}(x) = P(x) \wedge Q(x).$$

A rank-4 formula can witness at most four elements of one color inside a single E -class. Therefore, inside one equivalence class, color counts are visible only after truncation by $n \mapsto \min(n, 4)$.

Applying this to the six profiles gives

$$\tau_1 = (10, 0, 0, 0) \rightsquigarrow (4, 0, 0, 0), \quad \tau_2 = (6, 4, 0, 0) \rightsquigarrow (4, 4, 0, 0), \quad \tau_3 = (5, 5, 0, 0) \rightsquigarrow (4, 4, 0, 0),$$

$$\tau_4 = (4, 4, 2, 0) \rightsquigarrow (4, 4, 2, 0), \quad \tau_5 = (4, 4, 1, 1) \rightsquigarrow (4, 4, 1, 1), \quad \tau_6 = (3, 3, 2, 2) \rightsquigarrow (3, 3, 2, 2).$$

Thus τ_2 and τ_3 have the same rank-4 local behavior. The other four profiles remain locally distinguishable.

Step 2:

The profiles τ_1, τ_4, τ_6 occur exactly once in M_0 , and rank-4 logic can detect that uniqueness.

For τ_1 , use the class property $A(x) := \forall y (E(x, y) \rightarrow \chi_{00}(y))$. Then the sentence $\exists x (A(x) \wedge \forall y (A(y) \rightarrow E(x, y)))$ has quantifier rank at most 3 and says that there is exactly one E -class of type τ_1 .

For τ_4 , use $C(x) := \exists y (E(x, y) \wedge \chi_{01}(y)) \wedge \forall y (E(x, y) \rightarrow \neg \chi_{11}(y))$. This detects the unique profile with some 01-colored element and no 11-colored element. The analogous sentence $\exists x (C(x) \wedge \forall y (C(y) \rightarrow E(x, y)))$ has quantifier rank at most 3 and forces exactly one τ_4 -class.

For τ_6 , use $F(x) := \exists y \exists z (E(x, y) \wedge E(x, z) \wedge y \neq z \wedge \chi_{11}(y) \wedge \chi_{11}(z))$. This detects exactly the classes with at least two 11-colored elements, which among the allowed profiles means τ_6 . The sentence $\exists x (F(x) \wedge \forall y (F(y) \rightarrow E(x, y)))$ has quantifier rank at most 4 and forces exactly one τ_6 -class.

Therefore any $N \equiv_4 M_0$ must satisfy $n_1 = 1$, $n_4 = 1$, $n_6 = 1$, where $m(N) = (n_1, n_2, n_3, n_4, n_5, n_6)$.

Step 3:

Now consider the combined type formed by τ_2 and τ_3 . These two profiles both look locally like $(4, 4, 0, 0)$ to rank-4 formulas.

A representative property for this visible type is $B(x) := \exists y (E(x, y) \wedge \chi_{10}(y)) \wedge \forall y (E(x, y) \rightarrow (\neg \chi_{01}(y) \wedge \neg \chi_{11}(y)))$. This says that the E -class contains a 10-colored element but no 01- or 11-colored element. Among the allowed profiles, this is exactly the common visible type of τ_2 and τ_3 .

Rank 4 can force the existence of three such classes by the sentence

$$\exists x_1 \exists x_2 \exists x_3 \left(\bigwedge_{1 \leq i < j \leq 3} \neg E(x_i, x_j) \wedge \bigwedge_{i=1}^3 B(x_i) \right).$$

The three representatives use three quantifier levels, and the local test $B(x_i)$ uses one more level, so this fits within quantifier rank 4.

However, forcing four such classes would require four inequivalent representatives and still one additional quantifier level to verify the relevant local type. That exceeds rank 4. Equivalently, in the 4-round Ehrenfeucht–Fraïssé game, Spoiler cannot both mark four separate classes and also force a local verification of the combined τ_2/τ_3 profile in each of them.

Since M_0 has $2 + 3 = 5$ classes of type τ_2 or τ_3 , rank-4 logic only forces $n_2 + n_3 \geq 3$. It does not distinguish the split between n_2 and n_3 , and it does not force the combined number to be 4 or 5.

Step 4:

Next consider $\tau_5 = (4, 4, 1, 1)$. It is locally distinguishable from $\tau_6 = (3, 3, 2, 2)$ because a τ_5 -class has exactly one 11-colored element, while a τ_6 -class has two.

A useful representative property for τ_5 is $D(x) := \chi_{11}(x) \wedge \exists y (E(x, y) \wedge \chi_{01}(y)) \wedge \forall z ((E(x, z) \wedge \chi_{11}(z)) \rightarrow z = x)$. This says that x is the unique 11-colored element in its E -class and that the class also contains a 01-colored element. Among the allowed profiles, this detects τ_5 .

Rank 4 can force three τ_5 -classes using $\exists x_1 \exists x_2 \exists x_3 \left(\bigwedge_{1 \leq i < j \leq 3} \neg E(x_i, x_j) \wedge \bigwedge_{i=1}^3 D(x_i) \right)$. Again, the three class representatives use three quantifier levels, and the local test uses one more level.

But rank 4 cannot force four distinct τ_5 -classes, because that would require four inequivalent representatives together with a local check that those classes really have the τ_5 profile. Thus, since M_0 has four τ_5 -classes, the rank-4 consequence is exactly $n_5 \geq 3$.

Step 5:

Now use the fact that every structure in $\mathcal{K}$ has exactly 12 equivalence classes.

From Step 2, $n_1 = 1$, $n_4 = 1$, $n_6 = 1$. Therefore $n_2 + n_3 + n_5 = 9$.

From Steps 3 and 4, $n_2 + n_3 \geq 3$ and $n_5 \geq 3$. Hence $3 \leq n_2 + n_3 \leq 6$, and $n_5 = 9 - n_2 - n_3$.

So every possible vector must have the form $(1, n_2, n_3, 1, 9 - n_2 - n_3, 1)$, where $n_2, n_3 \in \mathbb{Z}_{\geq 0}$ and $3 \leq n_2 + n_3 \leq 6$.

Step 6:

It remains to justify that every vector satisfying these conditions really gives a structure rank-4 equivalent to M_0 .

The duplicator strategy in the 4-round Ehrenfeucht–Fraïssé game is this: preserve equality, preserve the E -relation between pebbled elements, and preserve the color-type of every pebbled element. If Spoiler plays inside an already pebbled E -class, Duplicator responds inside the corresponding E -class, using the matching truncated local color profile from Step 1. If Spoiler plays in a fresh E -class, Duplicator chooses a fresh class with the same visible role.

For the singleton visible types τ_1, τ_4, τ_6 , Step 2 shows that both structures have exactly one such class. For the combined τ_2/τ_3 visible type and for τ_5 , both structures have at least three relevant classes. In four rounds, any attempt to force the identity of a fresh class of one of these latter types must spend at least one additional move checking its local color pattern. Therefore Spoiler cannot force four independently verified classes of either type within rank 4.

Thus the above strategy preserves all atomic facts among the pebbled elements through four rounds, so $N \equiv_4 M_0$.

Step 7:

Combining the necessary conditions from Steps 2 through 5 with the sufficiency argument in Step 6 gives the complete set

$$\{(1, n_2, n_3, 1, 9 - n_2 - n_3, 1) : n_2, n_3 \in \mathbb{Z}_{\geq 0}, 3 \leq n_2 + n_3 \leq 6\}.$$

Final Answer:

$$\{(1, n_2, n_3, 1, 9 - n_2 - n_3, 1) : n_2, n_3 \in \mathbb{Z}_{\geq 0}, 3 \leq n_2 + n_3 \leq 6\}$$

Answer

$$\{(1, n_2, n_3, 1, 9 - n_2 - n_3, 1) : n_2, n_3 \in \mathbb{Z}_{\geq 0}, 3 \leq n_2 + n_3 \leq 6\}$$

Example Problem #3

Domain: Optimization and Numerical Mathematics

Sub-Domain: Approximation Theory

Problem Type: Optimization

Solution Type: Vector

PROBLEM PROMPT:

Let $\mathcal{P}_5$ be the real vector space of polynomials of degree at most 5. For $p \in \mathcal{P}_5$, define

$$\Lambda(p) = \max_{x \in X} w(x) |p(x) - y(x)|,$$

where $X = \{-5, -4, -2, -1, 0, 1, 3, 4, 6, 7, 9, 10\}$. The values of $w(x)$ and $y(x)$ are given by

x	-5	-4	-2	-1	0	1	3	4	6	7	9	10
$w(x)$	3	8	5	7	4	9	6	5	11	7	10	6
$y(x)$	$-\frac{722}{7}$	$-\frac{69613}{1344}$	$-\frac{373}{42}$	$-\frac{302}{105}$	$-\frac{69}{112}$	$\frac{37}{63}$	$\frac{6763}{1260}$	$\frac{1307}{105}$	$\frac{2883}{55}$	$\frac{10064}{105}$	$\frac{48044}{175}$	$\frac{9200}{21}$

Among all polynomials $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$ that minimize $\Lambda(p)$, determine

the coefficient vector $\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix}$.

DOMAIN JUSTIFICATION:

This problem involves finding the best weighted uniform approximation to finite data by a polynomial, which is part of Optimization and Numerical Mathematics, specifically Approximation theory. The problem also involves finite-dimensional linear algebra and exact rational arithmetic. However, those tools only support the main task: identifying the minimax approximant under a weighted sup-norm error. Therefore, Approximation theory is the best available subdomain.

STEP-BY-STEP SOLUTION & ANSWER:

Step-by-Step Solution

Step 1: Let

$$p_0(x) = -\frac{3}{7} + \frac{5}{4}x - \frac{2}{3}x^2 + \frac{7}{20}x^3 - \frac{1}{30}x^4 + \frac{1}{210}x^5.$$

We will show that this polynomial is the unique minimizer of $\Lambda(p)$. Define the weighted error

$$e_p(x) = w(x)(p(x) - y(x)).$$

Substituting p_0 into the data table gives the following weighted errors:

x	-5	-4	-2	-1	0	1	3	4	6	7	9	10
$e_{p_0}(x)$	1	$-\frac{7}{8}$	-1	1	$\frac{3}{4}$	-1	$-\frac{5}{6}$	1	$\frac{9}{10}$	-1	$-\frac{4}{5}$	1

Therefore $\Lambda(p_0) = 1$.

Step 2: The seven points

$$-5, -2, -1, 1, 4, 7, 10$$

have weighted errors

$$1, -1, 1, -1, 1, -1, 1.$$

At the remaining points $-4, 0, 3, 6, 9$, the absolute weighted errors are

$$\frac{7}{8}, \quad \frac{3}{4}, \quad \frac{5}{6}, \quad \frac{9}{10}, \quad \frac{4}{5},$$

all of which are strictly less than 1. Hence p_0 is feasible with maximum weighted error exactly 1.

Step 3: Let $q \in \mathcal{P}_5$ be any polynomial. Suppose first that $\Lambda(q) < 1$. At each point in (5), the weighted error of q satisfies $-1 < e_q(x) < 1$. Let $h(x) = q(x) - p_0(x)$. Since all weights are positive, $h(x)$ has the same sign as $e_q(x) - e_{p_0}(x)$ at every data point. Using (6) and (8), we get

$$h(-5) < 0, \quad h(-2) > 0, \quad h(-1) < 0, \quad h(1) > 0, \quad h(4) < 0, \quad h(7) > 0, \quad h(10) < 0.$$

Thus h changes sign on each of the six intervals $(-5, -2)$, $(-2, -1)$, $(-1, 1)$, $(1, 4)$, $(4, 7)$, $(7, 10)$. By the intermediate value theorem, h has at least one real root in each of those six disjoint intervals. Therefore h has at least six distinct real roots. But $h \in \mathcal{P}_5$, so this is impossible unless h is the zero polynomial. If $h \equiv 0$, then $q = p_0$, contradicting $\Lambda(q) < 1$ because $\Lambda(p_0) = 1$. Therefore no polynomial $q \in \mathcal{P}_5$ can satisfy $\Lambda(q) < 1$.

Step 4: Now suppose that $q \in \mathcal{P}_5$ is also a minimizer. Since Step 3 proves that no polynomial has maximum weighted error below 1, every minimizer must satisfy $\Lambda(q) = 1$. Again set $h = q - p_0$. At the seven points in (5), the bound $\Lambda(q) = 1$ gives $-1 \leq e_q(x) \leq 1$. Combining (6) and (12), we obtain

$$h(-5) \leq 0, \quad h(-2) \geq 0, \quad h(-1) \leq 0, \quad h(1) \geq 0, \quad h(4) \leq 0, \quad h(7) \geq 0, \quad h(10) \leq 0.$$

If one of the inequalities in (13) is strict, then either h changes sign across enough consecutive intervals to force six distinct roots, or else h vanishes at one of the listed points and still has roots in the adjacent sign-changing intervals. In every case, a nonzero $h \in \mathcal{P}_5$ would have at least six distinct real roots, which is impossible. Therefore all inequalities in (13) must be equalities, so

$$h(-5) = h(-2) = h(-1) = h(1) = h(4) = h(7) = h(10) = 0.$$

A polynomial of degree at most 5 with seven distinct roots must be the zero polynomial. Hence $h \equiv 0$, so $q = p_0$. Thus p_0 is the unique minimizer.

Step 5: From (1), the coefficients of the unique minimizing polynomial are

$$a_0 = -\frac{3}{7}, \quad a_1 = \frac{5}{4}, \quad a_2 = -\frac{2}{3}, \quad a_3 = \frac{7}{20}, \quad a_4 = -\frac{1}{30}, \quad a_5 = \frac{1}{210}.$$

Therefore the required coefficient vector is

Final Answer:

$$\begin{bmatrix} -\frac{3}{7} \\ \frac{5}{4} \\ -\frac{2}{3} \\ \frac{7}{20} \\ -\frac{1}{30} \\ \frac{1}{210} \end{bmatrix}$$

$$\begin{bmatrix} -\frac{3}{7} \\ \frac{5}{4} \\ -\frac{2}{3} \\ \frac{7}{20} \\ -\frac{1}{30} \\ \frac{1}{210} \end{bmatrix}$$

Example Problem #4

Domain: Geometry and Topology

Sub-Domain: Riemannian Geometry

Problem Type: Exact Computation

Solution Type: Matrix

PROBLEM PROMPT:

Let $M = \mathbb{R}^3$ have Riemannian metric $g = dx^2 + e^{2x}dy^2 + e^{2x+2y}dz^2$. Using the convention $\text{Ric}_{jk} = R^i_{jik}$ for

$$R^l_{ijk} = \partial_i \Gamma^l_{jk} - \partial_j \Gamma^l_{ik} + \Gamma^m_{jk} \Gamma^l_{im} - \Gamma^m_{ik} \Gamma^l_{jm},$$

compute the 3×3 matrix of $\text{Ric}^\sharp : T_p M \rightarrow T_p M$ at $p = (0, 0, 0)$ in the coordinate basis $(\partial_x, \partial_y, \partial_z)$.

DOMAIN JUSTIFICATION:

This problem involves a Riemannian metric on a smooth manifold, Christoffel symbols, the curvature tensor, the Ricci tensor, and the Ricci endomorphism, which are part of Geometry and Topology, Riemannian geometry.

STEP-BY-STEP SOLUTION & ANSWER:

Step-by-Step Solution

Step 1:

The metric components are $g_{xx} = 1$, $g_{yy} = e^{2x}$, $g_{zz} = e^{2x+2y}$, with all mixed components equal to 0. The inverse metric components are $g^{xx} = 1$, $g^{yy} = e^{-2x}$, $g^{zz} = e^{-2x-2y}$. The needed derivatives are $\partial_x g_{yy} = 2e^{2x}$, $\partial_x g_{zz} = 2e^{2x+2y}$, and $\partial_y g_{zz} = 2e^{2x+2y}$.

Step 2:

Using $\Gamma_{jk}^l = \frac{1}{2}g^{lm}(\partial_j g_{mk} + \partial_k g_{mj} - \partial_m g_{jk})$, the nonzero Christoffel symbols are computed as $\Gamma_{yy}^x = \frac{1}{2}g^{xx}(-\partial_x g_{yy}) = \frac{1}{2}(-2e^{2x}) = -e^{2x}$, $\Gamma_{zz}^x = \frac{1}{2}g^{xx}(-\partial_x g_{zz}) = \frac{1}{2}(-2e^{2x+2y}) = -e^{2x+2y}$, $\Gamma_{xy}^y = \Gamma_{yx}^y = \frac{1}{2}g^{yy}\partial_x g_{yy} = \frac{1}{2}e^{-2x}(2e^{2x}) = 1$, $\Gamma_{zz}^y = \frac{1}{2}g^{yy}(-\partial_y g_{zz}) = \frac{1}{2}e^{-2x}(-2e^{2x+2y}) = -e^{2y}$, $\Gamma_{xz}^z = \Gamma_{zx}^z = \frac{1}{2}g^{zz}\partial_x g_{zz} = \frac{1}{2}e^{-2x-2y}(2e^{2x+2y}) = 1$, and $\Gamma_{yz}^z = \Gamma_{zy}^z = \frac{1}{2}g^{zz}\partial_y g_{zz} = \frac{1}{2}e^{-2x-2y}(2e^{2x+2y}) = 1$. All remaining Christoffel symbols are 0.

Step 3:

The convention gives $\text{Ric}_{jk} = R_{jik}^i = \sum_i (\partial_j \Gamma_{ik}^i - \partial_i \Gamma_{jk}^i + \Gamma_{ik}^m \Gamma_{jm}^i - \Gamma_{jk}^m \Gamma_{im}^i)$. For Ric_{xx} , this becomes $\text{Ric}_{xx} = \sum_i \partial_x \Gamma_{ix}^i - \sum_i \partial_i \Gamma_{xx}^i + \sum_{i,m} \Gamma_{ix}^m \Gamma_{xm}^i - \sum_{i,m} \Gamma_{xx}^m \Gamma_{im}^i$. The first sum is $\partial_x \Gamma_{xx}^x = \partial_x \Gamma_{yy}^y + \partial_x \Gamma_{zz}^z = 0 + 0 + 0 = 0$. The second sum is 0 since $\Gamma_{xx}^x = \Gamma_{yy}^y = \Gamma_{zz}^z = 0$. The third sum is $\Gamma_{yx}^y \Gamma_{xy}^x + \Gamma_{zx}^z \Gamma_{xz}^z = 1 \cdot 1 + 1 \cdot 1 = 2$. The last sum is 0 since $\Gamma_{xx}^x = \Gamma_{yy}^y = \Gamma_{zz}^z = 0$. This gives $\text{Ric}_{xx} = 0 - 0 + 2 - 0 = 2$.

Step 4:

For Ric_{yy} , use $\text{Ric}_{yy} = \sum_i \partial_y \Gamma_{iy}^i - \sum_i \partial_i \Gamma_{yy}^i + \sum_{i,m} \Gamma_{iy}^m \Gamma_{ym}^i - \sum_{i,m} \Gamma_{yy}^m \Gamma_{im}^i$. The first sum is $\partial_y \Gamma_{xy}^x + \partial_y \Gamma_{yy}^y + \partial_y \Gamma_{zy}^z = 0 + 0 + 0 = 0$. The second sum is $-\partial_x \Gamma_{yy}^x - \partial_y \Gamma_{yy}^y - \partial_z \Gamma_{yy}^z = -\partial_x(-e^{2x}) - 0 - 0 = 2e^{2x}$. The third sum is $\Gamma_{xy}^y \Gamma_{yx}^x + \Gamma_{yy}^x \Gamma_{xy}^x + \Gamma_{zy}^z \Gamma_{yz}^z = 1(-e^{2x}) + (-e^{2x})(1) + 1 \cdot 1 = -2e^{2x} + 1$. The last sum is $-\Gamma_{yy}^x (\Gamma_{xx}^x + \Gamma_{yy}^y + \Gamma_{zz}^z) = -(-e^{2x})(0 + 1 + 1) = 2e^{2x}$. Therefore $\text{Ric}_{yy} = 0 + 2e^{2x} + (-2e^{2x} + 1) + 2e^{2x} = 2e^{2x} + 1$.

Step 5:

For Ric_{zz} , use $\text{Ric}_{zz} = \sum_i \partial_z \Gamma_{iz}^i - \sum_i \partial_i \Gamma_{zz}^i + \sum_{i,m} \Gamma_{iz}^m \Gamma_{zm}^i - \sum_{i,m} \Gamma_{zz}^m \Gamma_{im}^i$. The first sum is 0. The second sum is $-\partial_x(-e^{2x+2y}) - \partial_y(-e^{2y}) - \partial_z(0) = 2e^{2x+2y} + 2e^{2y}$. The third sum is $\Gamma_{xz}^x \Gamma_{zx}^x + \Gamma_{yz}^y \Gamma_{zy}^y + \Gamma_{zz}^x \Gamma_{xz}^x + \Gamma_{zz}^y \Gamma_{yz}^y = 1(-e^{2x+2y}) + 1(-e^{2y}) + (-e^{2x+2y})(1) + (-e^{2y})(1) = -2e^{2x+2y} - 2e^{2y}$. The last sum is $-\Gamma_{zz}^x (\Gamma_{xx}^x + \Gamma_{yy}^y + \Gamma_{zz}^z) - \Gamma_{zz}^y (\Gamma_{xx}^x + \Gamma_{yy}^y + \Gamma_{zz}^z) = -(-e^{2x+2y})(0 + 1 + 1) - (-e^{2y})(0 + 0 + 1) = 2e^{2x+2y} + e^{2y}$. Therefore $\text{Ric}_{zz} = 2e^{2x+2y} + e^{2y} = (2e^{2x} + 1)e^{2y}$.

Step 6:

For the off diagonal entries, substitution into $\text{Ric}_{jk} = \sum_i (\partial_j \Gamma_{ik}^i - \partial_i \Gamma_{jk}^i + \Gamma_{ik}^m \Gamma_{jm}^i - \Gamma_{jk}^m \Gamma_{im}^i)$ gives zero in each case, but the cancellations must be checked using the actual nonzero Christoffel symbols.

For Ric_{xy} , the derivative terms vanish: $\sum_i \partial_x \Gamma_{iy}^i = 0$, $\sum_i \partial_i \Gamma_{xy}^i = 0$. The product terms are $\sum_{i,m} \Gamma_{iy}^m \Gamma_{xm}^i = \Gamma_{zy}^z \Gamma_{xz}^z = 1$ and $\sum_{i,m} \Gamma_{xy}^m \Gamma_{im}^i = \Gamma_{xy}^y (\Gamma_{zy}^z) = 1$. Hence $\text{Ric}_{xy} = 0 - 0 + 1 - 1 = 0$. Similarly, for Ric_{yx} , the derivative terms vanish and $\sum_{i,m} \Gamma_{ix}^m \Gamma_{ym}^i = \Gamma_{zx}^z \Gamma_{yz}^z = 1$, while $\sum_{i,m} \Gamma_{yx}^m \Gamma_{im}^i = \Gamma_{yx}^y (\Gamma_{zy}^z) = 1$. Thus $\text{Ric}_{yx} = 0$. For the xz and zx entries, the only possible nonzero product contributions cancel in the same way, giving $\text{Ric}_{xz} = \text{Ric}_{zx} = 0$. For the yz and zy entries, the possible nonzero derivative and product contributions also cancel, giving $\text{Ric}_{yz} = \text{Ric}_{zy} = 0$. Thus the Ricci tensor is

$$\text{Ric} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2e^{2x} + 1 & 0 \\ 0 & 0 & (2e^{2x} + 1)e^{2y} \end{pmatrix}.$$

Step 7:

At $p = (0, 0, 0)$, the metric matrix is $g_p = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ because $e^0 = 1$. Raising one index uses

$(\text{Ric}^\sharp)^i_j = g^{ik} \text{Ric}_{kj}$. Since $g^{ik}(p) = \delta^{ik}$, the Ricci endomorphism has the same matrix as the Ricci tensor at p . Using $\text{Ric}_{xx}(p) = 2$, $\text{Ric}_{yy}(p) = 2e^0 + 1 = 3$, and $\text{Ric}_{zz}(p) = (2e^0 + 1)e^0 = 3$, we get

$$\text{Ric}_p^\sharp = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Final Answer: $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

Answer

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Example Problem #5

Domain: Number Theory

Sub-Domain: Elementary Number Theory

Problem Type: Exact Computation

Solution Type: Set or Multiset of Objects

PROBLEM PROMPT:

Let

$$g_k(x) = \frac{x^2 + 4x + k}{x^2 - 2x - 8}.$$

An integer x is called admissible if $-75 \leq x \leq 75$, x is not divisible by 3, $x \not\equiv 2 \pmod{5}$, and $x \neq -2, 4$.

We say that an integer k is balanced if there are exactly four admissible integers x for which $g_k(x)$ is an integer n satisfying $1 < g_k(x) < 14$, and among those four admissible values of x , the corresponding four integer outputs $g_k(x)$ contain exactly three distinct values.

Determine the complete set of balanced integers k .

DOMAIN JUSTIFICATION:

This problem involves integer-valued expressions, divisibility restrictions, and congruence conditions on the admissible inputs. The solver must analyze how integer solutions arise from the rational expression and determine which parameter values satisfy a prescribed multiplicity condition. The core reasoning involves modular arithmetic, divisibility, and the behavior of integers under algebraic constraints, which are central topics in Elementary Number Theory. Although the problem is expressed using a rational function, the primary focus is on arithmetic properties of integers rather than on algebraic manipulation alone.

STEP-BY-STEP SOLUTION & ANSWER:

Step-by-Step Solution

Step 1: Since $g_k(x)$ must be an integer satisfying $1 < g_k(x) < 14$, the possible integer outputs are $n \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$.

Step 2: Set $g_k(x) = n$. Then $\frac{x^2+4x+k}{x^2-2x-8} = n$. Solving for k gives $k = (n-1)x^2 - 2(n+2)x - 8n$.

Step 3: For each admissible integer x and each value $n \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$, evaluate $k = (n-1)x^2 - 2(n+2)x - 8n$. The values of k that occur exactly four times are 868, 1264, 3244, and 15844. The corresponding pairs are: for 868, $(-26, 2)$, $(-11, 7)$, $(14, 7)$, and $(34, 2)$; for 1264, $(-32, 2)$, $(13, 11)$, $(28, 3)$, and $(40, 2)$; for 3244, $(-22, 7)$, $(-17, 11)$, $(25, 7)$, and $(43, 3)$; and for 15844, $(-50, 7)$, $(46, 9)$, $(53, 7)$, and $(58, 6)$.

Step 4: A balanced value of k must occur exactly four times and produce exactly three distinct output values. The value 868 is excluded because its outputs are $\{2, 7, 7, 2\}$, which contain only two distinct values. The values 1264, 3244, and 15844 each produce exactly three distinct output values.

Step 5: Therefore, the complete set of balanced integers is $\{1264, 3244, 15844\}$. Final Answer:

$\{1264, 3244, 15844\}$

Answer

$\{1264, 3244, 15844\}$

Example Problem #6

Domain: Differential Equations and Dynamical Systems

Sub-Domain: Bifurcation Theory

Problem Type: Transformation Between Representations

Solution Type: Polynomial or Rational Function

PROBLEM PROMPT:

For real parameters ν, λ, μ , consider the autonomous system on $\mathbb{R}^4$ with coordinates (x, y, u, v) :

$$\begin{aligned}\dot{x} &= y + u + xu, \\ \dot{y} &= v + \mu + \lambda x + \left(\nu - \frac{2}{3}\right)x^2 + xy + 2y^2 + uv, \\ \dot{u} &= -2u + x^2 + 3xy - y^2 + xu + yv, \\ \dot{v} &= -3v + 2x^2 - xy + 4y^2 + xv - yu.\end{aligned}$$

For each sufficiently small real x , there is a unique formal solution $y(x), u(x), v(x)$ of the first, third, and fourth equilibrium equations with $y(x) = O(x^2)$, $u(x) = O(x^2)$, $v(x) = O(x^2)$. Let $J(x, \nu, \lambda, \mu)$ be the Jacobian of the full vector field with respect to (x, y, u, v) , evaluated at $(x, y(x), u(x), v(x))$. For sufficiently small real ν , there is a unique formal branch $x(\nu), \lambda(\nu), \mu(\nu)$ satisfying $x(\nu) = O(\nu)$, $\lambda(\nu) = O(\nu^2)$, $\mu(\nu) = O(\nu^3)$, such that the full system has an equilibrium at $(x(\nu), y(x(\nu)), u(x(\nu)), v(x(\nu)))$ and

$$\begin{aligned}\det J(x(\nu), \nu, \lambda(\nu), \mu(\nu)) &= 0, \\ \frac{d}{dx} \det J(x, \nu, \lambda(\nu), \mu(\nu)) \Big|_{x=x(\nu)} &= 0.\end{aligned}$$

Instead of observing the unfolding in the coordinates (λ, μ) , introduce the nonlinear parameter chart

$$\begin{aligned}L &= \lambda - \frac{5}{2}\lambda^2 + 3\mu + \frac{11}{4}\lambda\mu, \\ M &= \mu + \frac{7}{3}\lambda^2 - 4\lambda\mu - 2\lambda^3 + \frac{5}{6}\mu^2.\end{aligned}$$

Evaluate these expressions on the formal branch above, giving a formal curve $(L(\nu), M(\nu))$ in the observed parameter plane. Assign weighted degrees $\deg L = 2$, $\deg M = 3$. The defining ideal of this projected formal curve is generated by a unique monic Weierstrass series $Q(L, M) = M^2 + \tilde{A}(L)M + \tilde{B}(L)$, where every monomial of Q has weighted degree at least 6. Determine the exact weighted Taylor polynomial $Q_{\leq 14}(L, M)$ formed by all terms of weighted degree at most 14 in $Q(L, M)$.

DOMAIN JUSTIFICATION:

This problem involves local cusp bifurcations of nilpotent equilibria and formal reductions of autonomous systems, which are part of Differential Equations and Dynamical Systems and Bifurcation theory. The problem also involves formal power series elimination and polynomial reconstruction, which are part of algebra and symbolic computation. However, those calculations mainly serve to express the bifurcation set and are less central than the local dynamical structure.

STEP-BY-STEP SOLUTION & ANSWER:

Step-by-Step Solution

Step 1: Reduce the equilibrium and degeneracy conditions to one scalar equation

Let $F = (F_1, F_2, F_3, F_4)$ be the four right hand sides, and put $G = (F_1, F_3, F_4)$, $z = (y, u, v)$. At the origin, $\det \frac{\partial G}{\partial z}(0) = 6 \neq 0$. The formal implicit function theorem gives a unique formal graph $z = z(x) = (y(x), u(x), v(x))$ solving $G = 0$ and tangent to $z = 0$. Define $h(x, \nu, \lambda, \mu) = F_2(x, y(x), u(x), v(x))$. On this graph, the Schur complement gives

$$\det \frac{\partial F}{\partial(x, y, u, v)} = U(x) (F_{2,x} - F_{2,z} G_z^{-1} G_x) = U(x) h_x,$$

where $U(0) \neq 0$ is a formal unit. Since multiplication by a unit does not change the zero order, the requested branch is determined by $h = 0$, $h_x = 0$, $h_{xx} = 0$.

Step 2: Compute the reduced scalar series

Write $h(x, \nu, \lambda, \mu) = \mu + \lambda x + \nu x^2 + \sum_{k \geq 3} r_k x^k$. The first equilibrium equation gives $y = -(1+x)u$. With $u = \sum_{m \geq 2} u_m x^m$, $v = \sum_{m \geq 2} v_m x^m$, the coefficient equations $[x^m]F_3 = 0$ and $[x^m]F_4 = 0$ determine u_m and v_m recursively, because the new linear terms are $-2u_m$ and $-3v_m$. The coefficients needed through the requested weight are

k	3	4	5	6	7	8	9	10	11	12	13	14
r_k	$-\frac{1}{9}$	$\frac{149}{108}$	$\frac{53}{81}$	$-\frac{8819}{1944}$	$\frac{3821}{2916}$	$\frac{1505969}{139968}$	$-\frac{1028167}{104976}$	$-\frac{48496223}{2519424}$	$\frac{129322781}{3779136}$	$\frac{1950514525}{90699264}$	$-\frac{6163573997}{68024448}$	$\frac{30741316913}{1632586752}$

Step 3: Parametrize the cusp branch and pass to the observed coordinates

The equation $h_{xx} = 0$ has the form $2\nu + \sum_{k \geq 3} k(k-1)r_k x^{k-2} = 0$. Since the coefficient of x is $6r_3 = -\frac{2}{3}$, this equation has a unique formal solution $x = x(\nu) = O(\nu)$. Then $h_x = 0$ and $h = 0$ give

$$\lambda = -2\nu x - \sum_{k \geq 3} k r_k x^{k-1}, \quad \mu = -\lambda x - \nu x^2 - \sum_{k \geq 3} r_k x^k.$$

Substitution into $L = \lambda - \frac{5}{2}\lambda^2 + 3\mu + \frac{11}{4}\lambda\mu$ and $M = \mu + \frac{7}{3}\lambda^2 - 4\lambda\mu - 2\lambda^3 + \frac{5}{6}\mu^2$ gives

$$\begin{aligned} L(\nu) = & -3\nu^2 - 140\nu^3 - \frac{31865}{2}\nu^4 - \frac{4807625}{2}\nu^5 - \frac{20308982527}{48}\nu^6 \\ & - \frac{3939308931425}{48}\nu^7 - \frac{3268621431267977}{192}\nu^8 - \frac{6410425819397808019}{1728}\nu^9 \\ & - \frac{1450625269207089725639}{1728}\nu^{10} - \frac{338210748428744889866441}{1728}\nu^{11} \\ & - \frac{969429286187638907119932061}{20736}\nu^{12} - \frac{472494360282237715578175351759}{41472}\nu^{13} \\ & - \frac{234207369448159782914443034354657}{82944}\nu^{14} + O(\nu^{15}), \end{aligned}$$

and

$$\begin{aligned} M(\nu) = & 3\nu^3 + \frac{1425}{4}\nu^4 + \frac{210841}{4}\nu^5 + \frac{55189415}{6}\nu^6 + \frac{21306225751}{12}\nu^7 \\ & + \frac{35247406451561}{96}\nu^8 + \frac{1916042574055597}{24}\nu^9 + \frac{10389434876581005775}{576}\nu^{10} \\ & + \frac{7257955695332295323819}{1728}\nu^{11} + \frac{20783709887069591312432995}{20736}\nu^{12} \\ & + \frac{316310160705994425084365713}{1296}\nu^{13} + \frac{2507006755634221846166707661375}{41472}\nu^{14} + O(\nu^{15}). \end{aligned}$$

Step 4: Determine the monic Weierstrass relation

The weight bound and the monic condition force

$$Q_{\leq 14}(L, M) = M^2 + a_2 L^2 M + a_3 L^3 M + a_4 L^4 M + a_5 L^5 M + b_3 L^3 + b_4 L^4 + b_5 L^5 + b_6 L^6 + b_7 L^7.$$

Requiring $Q_{\leq 14}(L(\nu), M(\nu)) = O(\nu^{15})$ gives a triangular linear system. After already determined coefficients are substituted into each new order, the deciding equations are

ν -order	new coefficient	linear equation	value
6	b_3	$9 - 27b_3 = 0$	$\frac{1}{3}$
7	a_2	$\frac{1755}{2} + 27a_2 = 0$	$-\frac{65}{2}$
8	b_4	$\frac{878079}{16} + 81b_4 = 0$	$-\frac{292693}{432}$
9	a_3	$\frac{30582145}{3} - 81a_3 = 0$	$\frac{30582145}{243}$
10	b_5	$\frac{5036939317}{4} - 243b_5 = 0$	$\frac{5036939317}{972}$
11	a_4	$\frac{39465015209789}{144} + 243a_4 = 0$	$-\frac{39465015209789}{34992}$
12	b_6	$\frac{105620014432523345}{2592} + 729b_6 = 0$	$-\frac{105620014432523345}{1889568}$
13	a_5	$\frac{16682667515579699389}{1728} - 729a_5 = 0$	$\frac{16682667515579699389}{1259712}$
14	b_7	$\frac{24590093267001262845433}{15552} - 2187b_7 = 0$	$\frac{24590093267001262845433}{34012224}$

These values give the weighted Taylor polynomial

$$Q_{\leq 14}(L, M) = \frac{24590093267001262845433}{34012224} L^7 - \frac{105620014432523345}{1889568} L^6 + \frac{16682667515579699389}{1259712} L^5 M + \frac{5036939317}{972} L^5 - \frac{39465015209789}{34992} L^4 M - \frac{292693}{432} L^4 + \frac{30582145}{243} L^3 M + \frac{1}{3} L^3 - \frac{65}{2} L^2 M + M^2.$$

Substituting the displayed series for $L(\nu)$ and $M(\nu)$ gives $Q_{\leq 14}(L(\nu), M(\nu)) = O(\nu^{15})$, and the leading cancellation $M(\nu)^2 + \frac{1}{3} L(\nu)^3 = 9\nu^6 + \frac{1}{3}(-27\nu^6) + O(\nu^7) = O(\nu^7)$ checks the cusp normalization at the first nonzero weight.

Final Answer:
$$\frac{24590093267001262845433}{34012224} L^7 - \frac{105620014432523345}{1889568} L^6 + \frac{16682667515579699389}{1259712} L^5 M + \frac{5036939317}{972} L^5 - \frac{39465015209789}{34992} L^4 M - \frac{292693}{432} L^4 + \frac{30582145}{243} L^3 M + \frac{1}{3} L^3 - \frac{65}{2} L^2 M + M^2$$

Answer

$$\frac{24590093267001262845433}{34012224} L^7 - \frac{105620014432523345}{1889568} L^6 + \frac{16682667515579699389}{1259712} L^5 M + \frac{5036939317}{972} L^5 - \frac{39465015209789}{34992} L^4 M - \frac{292693}{432} L^4 + \frac{30582145}{243} L^3 M + \frac{1}{3} L^3 - \frac{65}{2} L^2 M + M^2$$

Example Problem #7

Domain: Linear Algebra

Sub-Domain: Finite-Dimensional Linear Algebra

Problem Type: Optimization

Solution Type: Set or Multiset of Objects

PROBLEM PROMPT:

Let $\mathcal{M}$ be the set of all triples (A, B, C) consisting of 3×3 complex matrices such that $[A, B] = C$, $[B, C] = A$ and $[C, A] = B$, where $[X, Y] = XY - 2YX$. Let

$$p := \inf_{(A,B,C) \in \mathcal{M}} \operatorname{Re}(\operatorname{tr}(e^A + e^B + e^C)) \in \mathbb{R} \cup \{\pm\infty\}$$

and

$$q := \sup_{(A,B,C) \in \mathcal{M}} \operatorname{Re}(\operatorname{tr}(e^A + e^B + e^C)) \in \mathbb{R} \cup \{\pm\infty\}.$$

What is the multiset $\{p, q\}$.

DOMAIN JUSTIFICATION:

This problem involves triples of 3×3 complex matrices, matrix products, a modified commutator relation $[X, Y] = XY - 2YX$, traces, and matrix exponentials. These are part of Linear Algebra, specifically Finite-dimensional linear algebra, because the central objects are finite-dimensional complex matrices and algebraic relations among them. The problem also involves taking an infimum and supremum of a real-valued expression, which is part of Optimization and Numerical Mathematics, specifically Nonlinear optimization. However, the optimization aspect is not the main mathematical structure of the problem. The feasible set $\mathcal{M}$ is defined entirely by matrix equations, and understanding the expression $\operatorname{Re}(\operatorname{tr}(e^A + e^B + e^C))$ depends primarily on finite-dimensional matrix behavior. Therefore, Linear Algebra, Finite-dimensional linear algebra is more appropriate than Optimization and Numerical Mathematics, Nonlinear optimization.

STEP-BY-STEP SOLUTION & ANSWER:

Step-by-Step Solution

Step 1: Reduction to irreducible blocks.

Let $V = \mathbb{C}^3$. A subspace $W \subseteq V$ is invariant if $AW \subseteq W$, $BW \subseteq W$, and $CW \subseteq W$. Because V is finite-dimensional, repeatedly choosing a nonzero invariant subspace of least possible dimension gives a chain $0 = V_0 \subset V_1 \subset \dots \subset V_r = V$ whose successive quotients are irreducible for the three operators A, B, C .

In a basis adapted to this chain, A, B, C are simultaneously block upper triangular. The diagonal blocks again satisfy $AB - 2BA = C$, $BC - 2CB = A$, $CA - 2AC = B$, because the diagonal block of a product of block upper triangular matrices is the product of the corresponding diagonal blocks.

If X is block upper triangular with diagonal blocks $X_1, \dots, X_r$, then X^n has diagonal blocks $X_1^n, \dots, X_r^n$. Therefore e^X has diagonal blocks $e^{X_1}, \dots, e^{X_r}$, and $\operatorname{tr}(e^X) = \sum_{j=1}^r \operatorname{tr}(e^{X_j})$. Thus the required expression is the sum of the corresponding expressions for irreducible blocks of dimensions 1, 2, 3.

We shall also use the following elementary fact. Over $\mathbb{C}$, every matrix can be triangularized. If the eigenvalues of X are $\lambda_1, \dots, \lambda_d$ with algebraic multiplicity, then triangularizing X shows that e^X is triangular with diagonal entries $e^{\lambda_1}, \dots, e^{\lambda_d}$. Hence $\operatorname{tr}(e^X) = e^{\lambda_1} + \dots + e^{\lambda_d}$.

Step 2: The support rule.

From $C = AB - 2BA$, $B = CA - 2AC$, substitution gives $B = (AB - 2BA)A - 2A(AB - 2BA) = 5ABA - 2BA^2 - 2A^2B$. Similarly, $C = 5ACA - 2CA^2 - 2A^2C$. So, for $X = B$ and also for $X = C$, $X = 5AXA - 2XA^2 - 2A^2X$. Define $F(s, t) := 5st - 2s^2 - 2t^2 = (s - 2t)(t - 2s)$. If

$A = \text{diag}(a_1, \dots, a_d)$, then the i, j entry of this identity is $(F(a_i, a_j) - 1) X_{ij} = 0$. Thus B_{ij} and C_{ij} can be nonzero only when $F(a_i, a_j) = 1$. Also, from $C = AB - 2BA$, $C_{ij} = (a_i - 2a_j) B_{ij}$.

The same support conclusion holds between generalized eigenspaces. If V_s and V_t are generalized eigenspaces of A , then on the block $V_t \rightarrow V_s$ the operator $X \mapsto 5AXA - 2XA^2 - 2A^2X - X$ is multiplication by $F(s, t) - 1$ plus a nilpotent operator. Therefore, if $F(s, t) \neq 1$, the block must be zero.

Consequently, in an irreducible block, the graph whose vertices are the eigenvalues of A and whose edges join s and t when $F(s, t) = 1$ must be connected. Otherwise the direct sum of the generalized eigenspaces in one connected component would be a nonzero proper invariant subspace.

Finally, an entry inside an eigenspace with eigenvalue s can occur only when $F(s, s) = s^2 = 1$.

Step 3: In an irreducible block of dimension greater than 1, A is diagonalizable with distinct eigenvalues.

First suppose $A = \lambda I$ on a block of dimension d . Then $C = AB - 2BA = -\lambda B$, $B = CA - 2AC = -\lambda C = \lambda^2 B$. If $\lambda^2 \neq 1$, then $B = 0$, hence $C = 0$, and then $A = BC - 2CB = 0$. Every subspace is invariant, so irreducibility forces $d = 1$.

If $\lambda^2 = 1$, then $C = -\lambda B$, and $A = BC - 2CB = B(-\lambda B) - 2(-\lambda B)B = \lambda B^2$. Since $A = \lambda I$, this gives $B^2 = I$. Thus B is diagonalizable with eigenspaces invariant under A, B, C ; if B is scalar, every subspace is invariant. Again irreducibility forces $d = 1$.

Now suppose A has exactly one eigenvalue but is not scalar. Write $A = \lambda I + N$, where $N \neq 0$ is nilpotent. If $\lambda^2 \neq 1$, the generalized support statement in Step 2 gives $B = C = 0$, impossible because then $A = BC - 2CB = 0$. Hence $\lambda^2 = 1$.

For $X = B$ and $X = C$, expanding the identity from Step 2 with $A = \lambda I + N$ gives $\lambda(NX + XN) + 5NXN - 2XN^2 - 2N^2X = 0$. If $v \in \ker N$, applying this displayed equation to v gives $N(\lambda I - 2N)Xv = 0$. Since $\lambda I - 2N$ is invertible and commutes with N , its inverse preserves $\ker N$. Thus $Xv \in \ker N$. Therefore $\ker N$ is invariant under A, B, C . Since $N \neq 0$ is nilpotent, $\ker N$ is nonzero and proper. This contradicts irreducibility.

Next suppose A has a 2 by 2 Jordan block with eigenvalue λ and one further eigenvalue $\mu \neq \lambda$. Put $A = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \mu \end{pmatrix}$. Let $X = (x_{ij})$ be either B or C . Expanding the identity from Step 2 gives, among other entries, $(F(\lambda, \mu) - 1)x_{23} = 0$, $(F(\lambda, \mu) - 1)x_{31} = 0$, $(F(\lambda, \mu) - 1)x_{13} + (5\mu - 4\lambda)x_{23} = 0$, $(F(\lambda, \mu) - 1)x_{32} + (5\mu - 4\lambda)x_{31} = 0$, and $(\lambda^2 - 1)x_{21} = 0$, $(\lambda^2 - 1)x_{11} + \lambda x_{21} = 0$. If $F(\lambda, \mu) \neq 1$, then the four displayed equations kill all entries connecting the generalized λ -eigenspace and the μ -eigenspace, giving a proper invariant subspace. Hence irreducibility forces $F(\lambda, \mu) = 1$. The two displayed equations involving x_{21} give $x_{21} = 0$ whether or not $\lambda^2 = 1$. If $5\mu - 4\lambda \neq 0$, then the fourth displayed equation gives $x_{31} = 0$, so both B and C send the line $\mathbb{C}e_1$ into itself, contradiction. Thus an irreducible case would have to satisfy $5\mu = 4\lambda$. Combining these two conditions gives $\mu = \frac{4\lambda}{5}$, $\lambda^2 = \frac{25}{18}$. Then $\lambda^2 \neq 1$ and $\mu^2 \neq 1$. The entries inside the λ -generalized

block and inside the μ -eigenspace force $x_{11} = x_{12} = x_{21} = x_{22} = x_{33} = 0$. Thus every such X has the form $X = \begin{pmatrix} 0 & 0 & r \\ 0 & 0 & s \\ t & u & 0 \end{pmatrix}$. So write $B = \begin{pmatrix} 0 & 0 & a \\ 0 & 0 & b \\ c & d & 0 \end{pmatrix}$. Using $C = AB - 2BA$ and then $BC - 2CB = A$ gives $bc = -\frac{\lambda}{2}$, $ac + bd = -\frac{1}{2}$, $9\lambda(ac + bd) - 4\lambda + 25bc = 0$. Substituting the first two equations into the third gives $9\lambda(-\frac{1}{2}) - 4\lambda + 25(-\frac{\lambda}{2}) = -21\lambda = 0$, contradicting $\lambda^2 = \frac{25}{18}$. Hence this Jordan-block case is impossible.

It remains to exclude the case where A is diagonalizable with exactly two eigenvalues, one of multiplicity 2. Write $V = U \oplus W$, where $A|_U = \lambda I_U$, $A|_W = \mu I_W$, $\dim U = 2$, $\dim W =$

1, $\lambda \neq \mu$. Irreducibility forces $F(\lambda, \mu) = 1$.

If $\lambda^2 \neq 1$, the support rule gives no $U \rightarrow U$ block for B or C . Let $r : U \rightarrow W$ be the $W \leftarrow U$ row block of B . By $C = AB - 2BA$, the corresponding row block of C is a scalar multiple of r . Choose $0 \neq u_0 \in \ker r$. Then $\mathbb{C}u_0$ is invariant under A, B, C , contradiction. Therefore $\lambda^2 = 1$.

Let $\lambda = \varepsilon$, where $\varepsilon \in \{1, -1\}$. Since $F(\varepsilon, \mu) = 1$ and $\mu \neq \varepsilon$, we get $\mu = \frac{3\varepsilon}{2}$. Write $B = \begin{pmatrix} P & u \\ r & 0 \end{pmatrix}$ relative to $U \oplus W$, where P is 2 by 2, u is a column, and r is a row. Then $C = AB - 2BA = \begin{pmatrix} -\varepsilon P & -2\varepsilon u \\ -\frac{\varepsilon}{2}r & 0 \end{pmatrix}$. The equation $BC - 2CB = A$ gives $P^2 + \frac{7}{2}ur = I_2$, $ru = -\frac{3}{2}$. Thus $u \neq 0$ and $r \neq 0$. For $s \in U$, the displayed block equation shows that $P^2s = s$ if and only if $r(s) = 0$. Hence $\ker r = \ker (P^2 - I_2)$. Since P commutes with P^2 , the line $\ker r$ is P -invariant. It is therefore invariant under A, B, C , contradiction.

Thus every irreducible block of dimension 2 or 3 has A diagonalizable with distinct eigenvalues.

Step 4: The one-dimensional irreducible blocks.

In dimension 1, write the scalars as a, b, c . The equations become $-ab = c$, $-bc = a$, $-ca = b$. If one of a, b, c is zero, then all three are zero. If none is zero, then $a^2 = b^2 = c^2 = 1$ and $c = -ab$. Therefore the scalar solutions are $(0, 0, 0)$, and $(1, 1, -1)$, $(1, -1, 1)$, $(-1, 1, 1)$, $(-1, -1, -1)$. The corresponding contributions are 3 , $2e + e^{-1}$, $3e^{-1}$.

Step 5: The two-dimensional irreducible blocks.

Let the irreducible block have dimension 2. By Step 3, $A = \begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix}$, $x \neq y$. Irreducibility forces $F(x, y) = 1$. Since A has distinct eigenvalues, any A -invariant subspace is spanned by eigenvectors of A ; hence both off-diagonal directions must occur. Write $B = \begin{pmatrix} u & \alpha \\ \beta & v \end{pmatrix}$, $\alpha\beta \neq 0$, where $u = 0$ unless $x^2 = 1$, and $v = 0$ unless $y^2 = 1$. From $C = AB - 2BA = \begin{pmatrix} -xu & (x - 2y)\alpha \\ (y - 2x)\beta & -yv \end{pmatrix}$, the equation $BC - 2CB = A$ gives $\alpha\beta(5y - 4x) + u^2x = x$, $\alpha\beta(5x - 4y) + v^2y = y$, $u(3x - 2y) + v(-2x + 3y) = 0$. First assume neither x nor y is 1 or -1 . Then $u = v = 0$. The first two scalar equations imply $\frac{x}{5y - 4x} = \frac{y}{5x - 4y}$, so $x^2 = y^2$. Since $x \neq y$, $y = -x$. Then $F(x, -x) = -9x^2 = 1$, so $x^2 = -\frac{1}{9}$, $\alpha\beta = -\frac{1}{9}$. The characteristic polynomial of each of A, B, C is $t^2 + \frac{1}{9}$. Thus the contribution is $6 \cos(\frac{1}{3})$.

Now suppose one eigenvalue is 1 or -1 . After swapping the basis vectors if necessary, take $x = \varepsilon$, $\varepsilon \in \{1, -1\}$. Since $F(\varepsilon, y) = 1$ and $y \neq \varepsilon$, $y = \frac{3\varepsilon}{2}$. Then $v = 0$, and the three scalar equations give $\alpha\beta = -\frac{3}{2}$, $u^2 = \frac{25}{4}$. Write $u = \tau\frac{5}{2}$ with $\tau \in \{1, -1\}$. The characteristic polynomials are $\chi_A(t) = (t - \varepsilon)(t - \frac{3\varepsilon}{2})$, $\chi_B(t) = t^2 - ut + \frac{3}{2} = (t - \tau)(t - \frac{3\tau}{2})$, $\chi_C(t) = t^2 + \varepsilon ut + \frac{3}{2} = (t + \varepsilon\tau)(t + \frac{3\varepsilon\tau}{2})$. The three signs ε , τ , and $-\varepsilon\tau$ have product -1 . Therefore either all three matrices have eigenvalues $-1, -\frac{3}{2}$, or exactly two matrices have eigenvalues $1, \frac{3}{2}$ and

the remaining one has eigenvalues $-1, -\frac{3}{2}$. The real two-dimensional contributions are therefore $3(e^{-1} + e^{-\frac{3}{2}})$ and $2(e + e^{\frac{3}{2}}) + e^{-1} + e^{-\frac{3}{2}}$.

Step 6: The three-dimensional irreducible blocks.

Let the irreducible block have dimension 3. By Step 3, $A = \text{diag}(x, y, z)$, where x, y, z are distinct. The support graph from Step 2 must be connected. It cannot be a triangle: if $F(x, y) = F(y, z) = F(z, x) = 1$, then subtracting pairs gives $5x = 2y + 2z$, $5y = 2z + 2x$, $5z = 2x + 2y$, which forces $x = y = z$. Therefore the graph is a path. Relabel the eigenvalues so that the path is $x - y - z$.

Thus $F(x, y) = 1$, $F(y, z) = 1$. For fixed y , the equation $F(t, y) = 1$ is $2t^2 - 5yt + 2y^2 + 1 = 0$. Its two roots are x and z , so $x + z = \frac{5y}{2}$. Irreducibility forces both directions on both edges. Write

$$B = \begin{pmatrix} u & \alpha & 0 \\ \beta & v & \gamma \\ 0 & \delta & w \end{pmatrix}, \quad \alpha\beta\gamma\delta \neq 0, \text{ where } u = 0 \text{ unless } x^2 = 1, v = 0 \text{ unless } y^2 = 1, \text{ and } w = 0 \text{ unless } z^2 = 1.$$

Multiplying out $C = AB - 2BA$ and $BC - 2CB = A$ gives

$$\begin{cases} \alpha\beta(5y - 4x) + u^2x = x, \\ \alpha\beta(5x - 4y) + \gamma\delta(5z - 4y) + v^2y = y, \\ \gamma\delta(5y - 4z) + w^2z = z, \\ u(3x - 2y) + v(-2x + 3y) = 0, \\ v(3y - 2z) + w(-2y + 3z) = 0. \end{cases}$$

First assume none of x, y, z is 1 or -1 . Then $u = v = w = 0$. The first and third scalar equations give $\alpha\beta = \frac{x}{5y-4x}$, $\gamma\delta = \frac{z}{5y-4z}$. Substitute these into the middle scalar equation: $\frac{x(5x-4y)}{5y-4x} + \frac{z(5z-4y)}{5y-4z} - y = 0$. Using $z = \frac{5y}{2} - x$, the left side of this equation simplifies to $-\frac{21y}{4}$. Hence $y = 0$. Then $z = -x$, and $F(x, 0) = 1$ gives $x^2 = -\frac{1}{2}$. Also $\alpha\beta = \gamma\delta = -\frac{1}{4}$. The characteristic polynomial of each of A, B, C is $t(t^2 + \frac{1}{2})$. Thus the contribution is $3 + 6 \cos\left(\frac{1}{\sqrt{2}}\right)$.

Now suppose one of x, y, z is 1 or -1 . The middle eigenvalue y cannot be 1 or -1 , because if $y = \varepsilon \in \{1, -1\}$, then $F(t, \varepsilon) = 1$ has roots ε and $\frac{3\varepsilon}{2}$, so there are not two roots distinct from y . After reversing the path if necessary, assume $x = \varepsilon$, $\varepsilon \in \{1, -1\}$. Then $y = \frac{3\varepsilon}{2}$, $z = \frac{11\varepsilon}{4}$. Only u can be nonzero among u, v, w . The three diagonal scalar equations give $\alpha\beta = -\frac{425}{56}$, $\gamma\delta = -\frac{11}{14}$, $u^2 = \frac{441}{16}$. Write $u = \tau\frac{21}{4}$ with $\tau \in \{1, -1\}$. The characteristic polynomial of A is $\chi_A(t) = (t - \varepsilon)(t - \frac{3\varepsilon}{2})(t - \frac{11\varepsilon}{4})$. For B , $\chi_B(t) = t^3 - ut^2 - (\alpha\beta + \gamma\delta)t + u\gamma\delta = t^3 - ut^2 + \frac{469}{56}t - \frac{11u}{14}$, so $\chi_B(t) = (t - \tau)(t - \frac{3\tau}{2})(t - \frac{11\tau}{4})$. For C , the first diagonal entry is $-\varepsilon u$, and the two adjacent off-diagonal products are again $\alpha\beta$ and $\gamma\delta$. Hence $\chi_C(t) = (t + \varepsilon\tau)(t + \frac{3\varepsilon\tau}{2})(t + \frac{11\varepsilon\tau}{4})$. The three signs ε, τ , and $-\varepsilon\tau$ have product -1 . Therefore either all three matrices have eigenvalues $-1, -\frac{3}{2}, -\frac{11}{4}$, or exactly two matrices have eigenvalues $1, \frac{3}{2}, \frac{11}{4}$ and the remaining one has eigenvalues $-1, -\frac{3}{2}, -\frac{11}{4}$. The real three-dimensional contributions are therefore $3\left(e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}}\right)$ and $2\left(e + e^{\frac{3}{2}} + e^{\frac{11}{4}}\right) + e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}}$.

Step 7: Comparing the finite list of possible sums.

The possible irreducible block contributions are as follows.

dimension	contribution	decimal value
1	$3e^{-1}$	1.104
1	3	3.000
1	$2e + e^{-1}$	5.804

2	$3 \left(e^{-1} + e^{-\frac{3}{2}} \right)$	1.773
2	$6 \cos \left(\frac{1}{3} \right)$	5.670
2	$2 \left(e + e^{\frac{3}{2}} \right) + e^{-1} + e^{-\frac{3}{2}}$	14.991
3	$3 \left(e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}} \right)$	1.965
3	$3 + 6 \cos \left(\frac{1}{\sqrt{2}} \right)$	7.561
3	$2 \left(e + e^{\frac{3}{2}} + e^{\frac{11}{4}} \right) + e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}}$	46.340

For total dimension 3, the possible partitions are $1 + 1 + 1$, $2 + 1$, and 3 .

For partition $1 + 1 + 1$, the least possible sum is $3 \cdot 3e^{-1} \approx 3.311$. For partition $2 + 1$, the least possible sum is $3 \left(e^{-1} + e^{-\frac{3}{2}} \right) + 3e^{-1} \approx 2.877$. For partition 3 , the least possible sum is $3 \left(e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}} \right) \approx 1.965$. Thus $p = 3 \left(e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}} \right)$.

For partition $1 + 1 + 1$, the greatest possible sum is $3(2e + e^{-1}) \approx 17.413$. For partition $2 + 1$, the greatest possible sum is $2 \left(e + e^{\frac{3}{2}} \right) + e^{-1} + e^{-\frac{3}{2}} + 2e + e^{-1} \approx 20.795$. For partition 3 , the greatest possible sum is $2 \left(e + e^{\frac{3}{2}} + e^{\frac{11}{4}} \right) + e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}} \approx 46.340$. Thus $q = 2 \left(e + e^{\frac{3}{2}} + e^{\frac{11}{4}} \right) + e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}}$. The exact comparison with the $2 + 1$ maximum is $q - \left[2 \left(e + e^{\frac{3}{2}} \right) + e^{-1} + e^{-\frac{3}{2}} + 2e + e^{-1} \right] = 2 \left(e^{\frac{11}{4}} - e \right) - \left(e^{-1} - e^{-\frac{11}{4}} \right) > 0$

Step 8: Explicit triples attaining the endpoint values and the final multiset.

For the minimum, take

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -\frac{3}{2} & 0 \\ 0 & 0 & -\frac{11}{4} \end{pmatrix}, \quad B = \begin{pmatrix} -\frac{21}{4} & 1 & 0 \\ -\frac{425}{56} & 0 & 1 \\ 0 & -\frac{11}{14} & 0 \end{pmatrix}, \quad C = \begin{pmatrix} -\frac{21}{4} & 2 & 0 \\ -\frac{425}{112} & 0 & 4 \\ 0 & -\frac{11}{56} & 0 \end{pmatrix}.$$

Direct multiplication verifies $AB - 2BA = C$, $BC - 2CB = A$, $CA - 2AC = B$. Each of A, B, C has characteristic polynomial $(t + 1) \left(t + \frac{3}{2} \right) \left(t + \frac{11}{4} \right)$, so this triple attains p .

$$\text{For the maximum, take } A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & \frac{11}{4} \end{pmatrix}, \quad B = \begin{pmatrix} \frac{21}{4} & 1 & 0 \\ -\frac{425}{56} & 0 & 1 \\ 0 & -\frac{11}{14} & 0 \end{pmatrix}, \quad C = \begin{pmatrix} -\frac{21}{4} & -2 & 0 \\ \frac{425}{112} & 0 & -4 \\ 0 & \frac{11}{56} & 0 \end{pmatrix}.$$

Again, $AB - 2BA = C$, $BC - 2CB = A$, $CA - 2AC = B$. Here A and B have characteristic polynomial $(t - 1) \left(t - \frac{3}{2} \right) \left(t - \frac{11}{4} \right)$, while C has characteristic polynomial $(t + 1) \left(t + \frac{3}{2} \right) \left(t + \frac{11}{4} \right)$, so this triple attains q .

Therefore the requested multiset is $\left\{ 3 \left(e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}} \right), 2 \left(e + e^{\frac{3}{2}} + e^{\frac{11}{4}} \right) + e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}} \right\}$

$$\text{Final Answer: } \left\{ 3 \left(e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}} \right), 2 \left(e + e^{\frac{3}{2}} + e^{\frac{11}{4}} \right) + e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}} \right\}$$

Answer

$$\left\{ 3 \left(e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}} \right), 2 \left(e + e^{\frac{3}{2}} + e^{\frac{11}{4}} \right) + e^{-1} + e^{-\frac{3}{2}} + e^{-\frac{11}{4}} \right\}$$

Example Problem #8

Domain: Abstract Algebra

Sub-Domain: Group Theory

Problem Type: Solve for Unknowns

Solution Type: Canonical Form

PROBLEM PROMPT:

Let $\mathbb{F}_2$ denote the field with two elements. Determine the number of matrices $M \in M_6(\mathbb{F}_2)$ (that is, 6×6 matrices with entries in $\mathbb{F}_2$) whose characteristic polynomial equals

$$\chi_M(x) = x^6 + x^5 + x^3 + x + 1 \in \mathbb{F}_2[x].$$

Express the answer as a positive integer.

DOMAIN JUSTIFICATION:

This problem involves counting matrices over a finite field $\mathbb{F}_2$ with a prescribed characteristic polynomial, which reduces to a sum over conjugacy classes in the general linear group $GL_6(\mathbb{F}_2)$ and uses the orbit-stabiliser identity for the conjugation action. The fundamental objects are conjugacy classes and centralisers of group elements, which are part of Abstract Algebra, sub-domain Group theory. The problem also involves rational canonical forms and the structure theorem for finitely generated modules over a principal ideal domain, which place auxiliary work inside Linear Algebra. However, the question never asks about a linear map, an eigenvector, a determinant, or a decomposition of a fixed matrix; it asks for the cardinality of an orbit set under group action, which is fundamentally a group-theoretic count. The Linear Algebra results enter only because the orbit data is indexed by invariant factor types, so Group theory is the correct primary classification.

STEP-BY-STEP SOLUTION & ANSWER:

Step-by-Step Solution

Step 1. Factor the characteristic polynomial over $\mathbb{F}_2$. It has no root since $\chi(0) = 1$ and $\chi(1) = 1$, so no linear factor. Testing the unique irreducible quadratic $p(x) = x^2 + x + 1$: $(x^2 + x + 1)^2 = x^4 + x^2 + 1$ and $(x^4 + x^2 + 1)(x^2 + x + 1) = x^6 + x^5 + x^3 + x + 1$. Hence $\chi(x) = p(x)^3$ with p irreducible of degree 2.

Step 2. Recast as module counting. A matrix M with characteristic polynomial χ turns $V = \mathbb{F}_2^6$ into an $\mathbb{F}_2[x]$ -module via $x \cdot v = Mv$. Matrices are conjugate iff the modules are isomorphic, and the number of matrices in one conjugacy class is $|GL_6(\mathbb{F}_2)|/|\text{Aut}_{\mathbb{F}_2[x]}(V)|$. So the total count is $\sum_{\lambda} |GL_6(\mathbb{F}_2)|/|\text{Aut}(V_{\lambda})|$ over the isomorphism types.

Step 3. Enumerate the types. As $\chi = p^3$ with p irreducible of degree 2, V is a module over the discrete valuation ring $\mathbb{F}_2[x]_{(p)}$ with residue field of size $Q = 2^2 = 4$, determined by a partition λ of 3: $V_{\lambda} = \bigoplus_i \mathbb{F}_2[x]/(p^{\lambda_i})$. The partitions of 3 are (3) , $(2, 1)$, $(1, 1, 1)$.

Step 4. The ambient group order is

$$|GL_6(\mathbb{F}_2)| = \prod_{i=0}^5 (2^6 - 2^i) = 63 \cdot 62 \cdot 60 \cdot 56 \cdot 48 \cdot 32 = 20158709760.$$

Step 5. Centralizer orders over the residue field $\mathbb{F}_Q$, $Q = 4$, using $|\text{Aut}(V_{\lambda})| = Q^{\sum_i (\lambda_i)^2} \prod_i \prod_{k=1}^{m_i} (1 - Q^{-k})$ (λ' the conjugate partition, m_i the number of parts equal to i): $\lambda = (3)$ is cyclic with $|\text{Aut}| = Q^3 - Q^2 = 48$; $\lambda = (2, 1)$ gives $|\text{Aut}| = 576$; $\lambda = (1, 1, 1)$ is semisimple with $|\text{Aut}| = |GL_3(\mathbb{F}_4)| = (4^3 - 1)(4^3 - 4)(4^3 - 4^2) = 181440$.

Step 6. Divide and add:

$$\frac{20158709760}{48} + \frac{20158709760}{576} + \frac{20158709760}{181440} = 419973120 + 34997760 + 111104 = 455081984.$$

Final Answer: 455081984

Answer _____
455081984

Example Problem #9

Domain: Calculus

Sub-Domain: Single-Variable Calculus

Problem Type: Optimization

Solution Type: Vector

PROBLEM PROMPT:

For real numbers a, b define

$$\Phi(a, b) = \int_0^1 (1+x) |x^4 - ax - b| dx.$$

Let (a_0, b_0) be the unique minimizer. Compute

$$\left(a_0, b_0, \Phi(a_0, b_0), \int_0^1 x^2(1+x) |x^4 - a_0x - b_0| dx \right).$$

DOMAIN JUSTIFICATION:

This problem involves minimizing a weighted integral of the absolute value of a one-variable function, deriving first-order balance conditions under the integral sign, analyzing the sign pattern of a convex one-variable residual, and evaluating weighted one-variable definite integrals. These are part of Calculus, specifically Single-variable calculus. The problem also involves affine approximation and convex optimization ideas. However, those ideas are used only to organize the minimization, while the central work is one-variable sign analysis and exact integration, so Calculus is the best classification.

STEP-BY-STEP SOLUTION & ANSWER

Step-by-Step Solution

Step 1: Set up the residual and the weighted balance equations.

Let

$$R_{a,b}(x) = x^4 - ax - b.$$

Then

$$\Phi(a, b) = \int_0^1 (1+x) |R_{a,b}(x)| dx.$$

At a minimizing pair (a_0, b_0) , write

$$R(x) = R_{a_0, b_0}(x) = x^4 - a_0x - b_0.$$

Since R is a nonzero polynomial, its zero set has measure 0. Therefore differentiating Φ with respect to a and b gives

$$\frac{\partial \Phi}{\partial b} = - \int_0^1 (1+x) \operatorname{sgn}(R(x)) dx$$

and

$$\frac{\partial \Phi}{\partial a} = - \int_0^1 x(1+x) \operatorname{sgn}(R(x)) dx.$$

Thus the minimizing residual must satisfy

$$\int_0^1 (1+x) \operatorname{sgn}(R(x)) dx = 0$$

and

$$\int_0^1 x(1+x) \operatorname{sgn}(R(x)) dx = 0.$$

These are the two weighted balance equations.

Step 2: Determine the sign pattern of the minimizing residual.

The residual is

$$R(x) = x^4 - a_0x - b_0.$$

Its derivative is

$$R'(x) = 4x^3 - a_0.$$

Since $4x^3 - a_0$ is strictly increasing on $[0, 1]$, the function R can have at most two zeros in $[0, 1]$.

The residual cannot have no sign change, because then

$$\int_0^1 (1+x) \operatorname{sgn}(R(x)) dx$$

would be nonzero.

It also cannot have only one sign change. Indeed, suppose the sign changes once at c . Then the first balance equation would force

$$\int_0^c (1+x) dx = \frac{1}{2} \int_0^1 (1+x) dx.$$

Since

$$\int_0^1 (1+x) dx = \frac{3}{2},$$

this gives

$$c + \frac{c^2}{2} = \frac{3}{4}.$$

If the second balance equation also held, then we would need

$$\int_0^c x(1+x) dx = \frac{1}{2} \int_0^1 x(1+x) dx.$$

Since

$$\int_0^1 x(1+x) dx = \frac{5}{6},$$

this would give

$$\frac{c^2}{2} + \frac{c^3}{3} = \frac{5}{12}.$$

But from

$$c + \frac{c^2}{2} = \frac{3}{4}$$

we get

$$c^2 = \frac{3}{2} - 2c.$$

Then

$$c^3 = c \left(\frac{3}{2} - 2c \right) = \frac{3}{2}c - 2 \left(\frac{3}{2} - 2c \right) = \frac{11}{2}c - 3.$$

Hence

$$\frac{c^2}{2} + \frac{c^3}{3} = \frac{1}{2} \left(\frac{3}{2} - 2c \right) + \frac{1}{3} \left(\frac{11}{2}c - 3 \right) = \frac{5}{6}c - \frac{1}{4}.$$

Setting this equal to $\frac{5}{12}$ gives

$$\frac{5}{6}c - \frac{1}{4} = \frac{5}{12},$$

so

$$c = \frac{4}{5}.$$

But $c = \frac{4}{5}$ does not satisfy

$$c + \frac{c^2}{2} = \frac{3}{4}.$$

Therefore one sign change is impossible.

So the minimizing residual has exactly two zeros

$$0 < r < s < 1.$$

Because R' is increasing, the sign pattern must be

$$R(x) > 0 \quad \text{on } [0, r) \cup (s, 1],$$

and

$$R(x) < 0 \quad \text{on } (r, s).$$

Step 3: Convert the balance equations into equations for the two sign-change points.

From the sign pattern, the first balance equation says that the weighted mass of the middle interval is half of the total weighted mass:

$$\int_r^s (1+x) dx = \frac{1}{2} \int_0^1 (1+x) dx.$$

Since

$$\int_0^1 (1+x) dx = \frac{3}{2},$$

we get

$$\int_r^s (1+x) dx = \frac{3}{4}.$$

Thus

$$(s-r) + \frac{s^2 - r^2}{2} = \frac{3}{4}.$$

The second balance equation similarly gives

$$\int_r^s x(1+x) dx = \frac{1}{2} \int_0^1 x(1+x) dx.$$

Since

$$\int_0^1 x(1+x) dx = \frac{5}{6},$$

we get

$$\int_r^s x(1+x) dx = \frac{5}{12}.$$

Thus

$$\frac{s^2 - r^2}{2} + \frac{s^3 - r^3}{3} = \frac{5}{12}.$$

Now introduce

$$\alpha = r + s, \quad \delta = s - r.$$

Then

$$s^2 - r^2 = \delta\alpha$$

and

$$s^3 - r^3 = \delta(s^2 + sr + r^2).$$

Because

$$s^2 + sr + r^2 = \frac{3\alpha^2 + \delta^2}{4},$$

the two equations become

$$\delta \left(1 + \frac{\alpha}{2}\right) = \frac{3}{4}$$

and

$$\delta \left(\frac{\alpha}{2} + \frac{3\alpha^2 + \delta^2}{12}\right) = \frac{5}{12}.$$

Step 4: Solve for the intrinsic parameters α and δ .

From

$$\delta \left(1 + \frac{\alpha}{2}\right) = \frac{3}{4},$$

we get

$$\delta = \frac{3}{2(\alpha + 2)}.$$

Substitute this into

$$\delta \left(\frac{\alpha}{2} + \frac{3\alpha^2 + \delta^2}{12}\right) = \frac{5}{12}.$$

After clearing denominators, this gives

$$36\alpha^4 + 176\alpha^3 + 192\alpha^2 - 192\alpha - 293 = 0.$$

Let α be the unique root of

$$36t^4 + 176t^3 + 192t^2 - 192t - 293 = 0$$

in the interval

$$1 < t < \frac{6}{5}.$$

This uniqueness is clear because the derivative

$$144t^3 + 528t^2 + 384t - 192$$

is positive for $t \geq 1$, while the polynomial is negative at $t = 1$ and positive at $t = \frac{6}{5}$.

Then

$$\delta = \frac{3}{2(\alpha + 2)}.$$

The two sign-change points are

$$r = \frac{\alpha - \delta}{2}, \quad s = \frac{\alpha + \delta}{2}.$$

Step 5: Recover the minimizing parameters a_0 and b_0 .

Since $R(r) = R(s) = 0$, we have

$$r^4 - a_0 r - b_0 = 0$$

and

$$s^4 - a_0 s - b_0 = 0.$$

Subtracting these equations gives

$$a_0 = \frac{s^4 - r^4}{s - r}.$$

Therefore

$$a_0 = s^3 + s^2 r + s r^2 + r^3.$$

Using

$$\alpha = r + s, \quad \delta = s - r,$$

we get

$$a_0 = \frac{\alpha(\alpha^2 + \delta^2)}{2}.$$

Next,

$$b_0 = r^4 - a_0 r.$$

Substituting the expression for a_0 gives

$$b_0 = -rs(s^2 + sr + r^2).$$

Now

$$rs = \frac{\alpha^2 - \delta^2}{4}$$

and

$$s^2 + sr + r^2 = \frac{3\alpha^2 + \delta^2}{4}.$$

Hence

$$b_0 = -\frac{(\alpha^2 - \delta^2)(3\alpha^2 + \delta^2)}{16}.$$

Step 6: Express the needed signed moments.

For $n \geq 0$, define

$$A_n = \int_0^1 x^n \operatorname{sgn}(R(x)) dx.$$

Using the sign pattern $+, -, +$, we have

$$A_n = \int_0^r x^n dx - \int_r^s x^n dx + \int_s^1 x^n dx.$$

Thus

$$A_n = \frac{1}{n+1} - \frac{2(s^{n+1} - r^{n+1})}{n+1}.$$

Since

$$r = \frac{\alpha - \delta}{2}, \quad s = \frac{\alpha + \delta}{2},$$

this can be written as

$$A_n = \frac{1}{n+1} - \frac{2}{n+1} \left[\left(\frac{\alpha + \delta}{2} \right)^{n+1} - \left(\frac{\alpha - \delta}{2} \right)^{n+1} \right].$$

Step 7: Compute the minimum value and the weighted residual moment.

For $n \geq 0$, define

$$A_n = \frac{1}{n+1} - \frac{2}{n+1} \left[\left(\frac{\alpha + \delta}{2} \right)^{n+1} - \left(\frac{\alpha - \delta}{2} \right)^{n+1} \right],$$

where α is the unique root in $(1, \frac{6}{5})$ of

$$36t^4 + 176t^3 + 192t^2 - 192t - 293 = 0$$

and

$$\delta = \frac{3}{2(\alpha + 2)}.$$

From the previous steps,

$$a_0 = \frac{\alpha(\alpha^2 + \delta^2)}{2}$$

and

$$b_0 = -\frac{(\alpha^2 - \delta^2)(3\alpha^2 + \delta^2)}{16}.$$

The minimum value is

$$\Phi(a_0, b_0) = A_4 + A_5.$$

The weighted residual moment is

$$\int_0^1 x^2(1+x)|x^4 - a_0x - b_0| dx = A_6 + A_7 - a_0(A_3 + A_4) - b_0(A_2 + A_3).$$

Substituting the formulas for a_0 and b_0 , this becomes

$$A_6 + A_7 - \frac{\alpha(\alpha^2 + \delta^2)}{2}(A_3 + A_4) + \frac{(\alpha^2 - \delta^2)(3\alpha^2 + \delta^2)}{16}(A_2 + A_3).$$

Here the symbols used in the final answer are exactly the following: α is the unique root in $(1, \frac{6}{5})$ of $36t^4 + 176t^3 + 192t^2 - 192t - 293 = 0$, $\delta = \frac{3}{2(\alpha+2)}$, and

$$A_n = \frac{1}{n+1} - \frac{2}{n+1} \left[\left(\frac{\alpha + \delta}{2} \right)^{n+1} - \left(\frac{\alpha - \delta}{2} \right)^{n+1} \right].$$

Final Answer: $\left(\frac{\alpha(\alpha^2 + \delta^2)}{2}, -\frac{(\alpha^2 - \delta^2)(3\alpha^2 + \delta^2)}{16}, A_4 + A_5, A_6 + A_7 - \frac{\alpha(\alpha^2 + \delta^2)}{2}(A_3 + A_4) + \frac{(\alpha^2 - \delta^2)(3\alpha^2 + \delta^2)}{16}(A_2 + A_3) \right)$

Answer $\left(\frac{\alpha(\alpha^2 + \delta^2)}{2}, -\frac{(\alpha^2 - \delta^2)(3\alpha^2 + \delta^2)}{16}, A_4 + A_5, A_6 + A_7 - \frac{\alpha(\alpha^2 + \delta^2)}{2}(A_3 + A_4) + \frac{(\alpha^2 - \delta^2)(3\alpha^2 + \delta^2)}{16}(A_2 + A_3) \right)$

Example Problem #10

Domain: Differential Equations and Dynamical Systems

Sub-Domain: Partial Differential Equations

Problem Type: Symbolic Derivation

Solution Type: Set or Multiset of Objects

PROBLEM PROMPT:

Consider the inviscid Burgers equation on the real line,

$$u_t + \left(\frac{u^2}{2} \right)_x = 0, \quad x \in \mathbb{R}, t > 0,$$

and let $u(x, t)$ denote its entropy (vanishing-viscosity) solution, i.e. the unique bounded weak solution obtained as the $\nu \rightarrow 0^+$ limit of solutions of $u_t + uu_x = \nu u_{xx}$. Admissible discontinuities are exactly those satisfying the Lax entropy condition $u(x^-, t) > u(x^+, t)$ and propagating at the Rankine-Hugoniot speed $\frac{1}{2}(u(x^-, t) + u(x^+, t))$; everywhere else the solution is determined by the method of characteristics, with rarefaction waves filling any expansive corner.

Fix arbitrary real numbers $w_1, w_2, w_3 > 0$ (no relation among them is assumed). Put $a_1 = 0$ and $a_{k+1} = a_k + w_k$ for $k = 1, 2, 3$. The initial datum is the bounded piecewise-constant profile

$$u(x, 0) = \begin{cases} 4, & x < a_1, \\ 3, & a_1 \leq x < a_2, \\ 1, & a_2 \leq x < a_3, \\ 2, & a_3 \leq x < a_4, \\ 0, & x \geq a_4, \end{cases}$$

a descending cascade whose monotonicity is broken exactly once: the levels 2 and 1 appear in transposed order, so the profile steps $4 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 0$ across the breakpoints a_1, a_2, a_3, a_4 .

For each $t > 0$ let $n(t)$ be the number of points $x \in \mathbb{R}$ at which the map $x \mapsto u(x, t)$ is discontinuous. The function $t \mapsto n(t)$ is non-increasing and right-continuous, with $n(0^+) = 3$, and $n(t) = 1$ for all sufficiently large t . Let $\mathcal{T}$ be the multiset of discontinuity times of $n(\cdot)$, each such time τ counted with multiplicity $n(\tau^-) - n(\tau)$; thus $\mathcal{T}$ contains exactly 2 elements (with multiplicity), and knowing $\mathcal{T}$ is equivalent to knowing $n(t)$ for every $t > 0$.

Determine the multiset $\mathcal{T}$ as exact closed-form expressions in w_1, w_2, w_3 , valid for every choice of positive widths; where the form of an element of $\mathcal{T}$ depends on the widths, the case distinctions must be given by explicit inequalities in w_1, w_2, w_3 that partition all of $(0, \infty)^3$.

DOMAIN JUSTIFICATION:

This problem involves shock waves, a rarefaction wave, entropy admissibility, jump speed relations, and the timetable on which colliding wave fronts of the inviscid Burgers equation merge, which are part of Differential Equations and Dynamical Systems within Partial differential equations. The problem also involves a small linear ordinary differential equation for the shock paths inside the rarefaction plus collision time comparisons through explicit width inequalities, which are part of Ordinary differential equations and basic algebra. However, those pieces just feed the interaction bookkeeping of the conservation law, so the weight of the problem sits with the partial differential equation.

STEP-BY-STEP SOLUTION & ANSWER:

Step-by-Step Solution

Step 1: Set up the initial wave configuration and reduce the merger schedule to two collision times

Throughout, $a_1 = 0$, $a_2 = w_1$, $a_3 = w_1 + w_2$, $a_4 = w_1 + w_2 + w_3$, and $W = w_1 + 3w_2$. The ambient setting is the classical entropy (Kruzhkov, vanishing-viscosity) solution of $u_t + \left(\frac{u^2}{2}\right)_x = 0$ on $\mathbb{R} \times (0, \infty)$: bounded, with finitely many piecewise- C^1 discontinuity curves. All times appearing below are finite real numbers and every merger time is attained, because the colliding fronts are continuous curves with strictly ordered speeds, certified case by case.

Each initial jump resolves as a Riemann problem. At a_1 the drop $4 \rightarrow 3$ produces a Lax-admissible shock S_1 with Rankine-Hugoniot speed $\frac{4+3}{2} = \frac{7}{2}$, so $x_1(t) = \frac{7}{2}t$. The drop $3 \rightarrow 1$ at a_2 produces the shock S_2 with speed $\frac{3+1}{2} = 2$, so $x_2(t) = w_1 + 2t$. The lone increasing jump $1 \rightarrow 2$ at a_3 opens a rarefaction fan $u(x, t) = \frac{x-a_3}{t}$ on $a_3 + t \leq x \leq a_3 + 2t$. That fan is continuous, so it contributes no discontinuity. And at a_4 the drop $2 \rightarrow 0$ produces the shock S_3 with speed $\frac{2+0}{2} = 1$, so $x_3(t) = a_4 + t$.

So $n(0^+) = 3$. The final state is a single $4|0$ shock of speed 2, so exactly two mergers occur: either S_1 absorbs S_2 first and the merged front later absorbs S_3 , or S_2 meets S_3 first (through the fan) and S_1 later absorbs the result. Until the first merger, S_1 , S_2 , S_3 and the fan evolve without influencing one another, since each shock is driven only by its two adjacent states. That gives

$$\tau_1 = \min(t_L, t_R),$$

where t_L denotes the collision time of S_1 with the unperturbed S_2 , and t_R denotes the collision time of the unperturbed S_2 with the unperturbed S_3 through the fan.

Step 2: Derive the trajectory law for shocks inside the rarefaction fan

If a shock has constant outer state c on one side and fan values on the other, the Rankine-Hugoniot relation gives the ODE

$$\dot{x} = \frac{1}{2} \left(c + \frac{x - a_3}{t} \right).$$

Substituting $y = x - a_3$ turns this into the linear equation $\dot{y} = \frac{c}{2} + \frac{y}{2t}$. The homogeneous solution is $t^{1/2}$, and $y = ct$ works as a particular solution (check: $\frac{d}{dt}(ct) = c = \frac{c}{2} + \frac{ct}{2t}$). So every in-fan shock trajectory has the form

$$x(t) = a_3 + ct + C\sqrt{t},$$

with the constant C fixed by the entry point. The fan value carried at the shock is $v(t) = \frac{x-a_3}{t} = c + Ct^{-1/2}$. Each such shock stays Lax-admissible the whole way. The $c = 3$ front keeps fan values below 3 on its right, the $c = 0$ front keeps fan values above 0 on its left, and the merged front keeps 4 on the left against fan values below 4 on the right.

Step 3: Compute t_L , the time for S_1 to catch S_2

S_2 travels at speed 2 and reaches the fan's left edge $x = a_3 + t$ when $w_1 + 2t = a_3 + t$, that is at $t = w_2$, at $x = a_3 + w_2$. Inside the fan its trajectory follows the law $x(t) = a_3 + 3t + C\sqrt{t}$ (constant left state $c = 3$), and matching the entry point forces $w_2 = 3w_2 + C\sqrt{w_2}$, so $C = -2\sqrt{w_2}$:

which starts at $v_2(w_2) = 1$ (continuity) and increases. S_2 exhausts the fan ($v_2 = 2$) at $t = 4w_2$, at $x = a_3 + 8w_2$, which is exactly the fan's right edge $a_3 + 2t$ at that instant; afterwards (if the constant-2 band still exists, that is $4w_2 \leq w_3$) it proceeds as a straight 3|2 shock of speed $\frac{5}{2}$:

$$x_2(t) = a_3 + \frac{5}{2}t - 2w_2 \quad (t \geq 4w_2).$$

S_1 remains the straight line $\frac{7}{2}t$ until impact, and setting $\frac{7}{2}t$ equal to each branch of x_2 gives three phases. In the straight phase ($t \leq w_2$): $\frac{7}{2}t = w_1 + 2t$, so $t = \frac{2w_1}{3}$, valid iff $\frac{2w_1}{3} \leq w_2$, that is $2w_1 \leq 3w_2$. In the in-fan phase ($w_2 \leq t \leq 4w_2$): $\frac{7}{2}t = a_3 + 3t - 2\sqrt{w_2 t}$ rearranges to $\frac{t}{2} + 2\sqrt{w_2 t} = w_1 + w_2$; with $s = \sqrt{t}$ this reads $s^2 + 4\sqrt{w_2} s - 2(w_1 + w_2) = 0$, whose positive root is $s = \sqrt{2w_1 + 6w_2} - 2\sqrt{w_2}$ (the discriminant is $16w_2 + 8(w_1 + w_2) = 4(2w_1 + 6w_2)$), so

$$t_L = (\sqrt{2w_1 + 6w_2} - 2\sqrt{w_2})^2,$$

valid on $w_2 \leq t_L \leq 4w_2$: indeed $t_L \geq w_2 \iff \sqrt{2w_1 + 6w_2} \geq 3\sqrt{w_2} \iff 2w_1 \geq 3w_2$, and $t_L \leq 4w_2 \iff \sqrt{2w_1 + 6w_2} \leq 4\sqrt{w_2} \iff w_1 \leq 5w_2$. In the post-fan phase ($t \geq 4w_2$): $\frac{7}{2}t = a_3 + \frac{5}{2}t - 2w_2$, so $t = w_1 - w_2$, valid iff $w_1 - w_2 \geq 4w_2$, that is $w_1 \geq 5w_2$.

The three validity windows about exactly at $2w_1 = 3w_2$ and $w_1 = 5w_2$, with matching values w_2 and $4w_2$ respectively (at $w_1 = \frac{3}{2}w_2$, $(3\sqrt{w_2} - 2\sqrt{w_2})^2 = w_2$; at $w_1 = 5w_2$, $(4\sqrt{w_2} - 2\sqrt{w_2})^2 = 4w_2$). The post-fan phase presumes the fan survives until $4w_2$; if instead S_3 reaches S_2 earlier, that collision obeys $t_R \leq 4w_2$, while no catch by S_1 can occur before $4w_2$ when $w_1 \geq 5w_2$ (the validity windows just derived place every catch at $t \geq 4w_2$ in that range), so the minimum defining τ_1 is unaffected.

Step 4: Compute t_R , the time for S_2 to meet S_3 through the fan

S_3 travels at speed 1 until the fan's right edge $a_3 + 2t$ overtakes it at $t = w_3$, at $x = a_4 + w_3 = a_3 + 2w_3$. Inside the fan its trajectory follows the law $x(t) = a_3 + C\sqrt{t}$ (constant right state $c = 0$), and matching the entry point gives $2w_3 = C\sqrt{w_3}$, so $C = 2\sqrt{w_3}$:

$$x_3(t) = a_3 + 2\sqrt{w_3 t}, \quad v_3(t) = 2\sqrt{w_3/t} \quad (t \geq w_3),$$

starting at $v_3(w_3) = 2$ (continuity) and decreasing. S_3 exhausts the fan ($v_3 = 1$, possible only if the value-1 left edge $x = a_3 + t$ is still present, that is $4w_3 \leq w_2$) at $t = 4w_3$, at $x = a_3 + 4w_3$, exactly the left edge at that instant; afterwards it proceeds as a straight 1|0 shock of speed $\frac{1}{2}$:

$$x_3(t) = a_3 + \frac{t}{2} + 2w_3 \quad (t \geq 4w_3).$$

The collision happens in one of three mutually exclusive modes, and the orders are forced by the geometry: a straight S_2 cannot touch S_3 while the fan's value-1 edge $x = a_3 + t$ lies between them (it must cross that characteristic first, at $t = w_2$), and symmetrically a straight S_3 cannot be touched from the left while the value-2 edge $x = a_3 + 2t$ separates it (that edge arrives only at $t = w_3$).

Mode 1, $w_3 \geq 4w_2$ (S_2 exhausts the fan first, at $4w_2 \leq w_3$): straight 3|2 against straight 2|0: $a_3 + \frac{5}{2}t - 2w_2 = a_4 + t$ gives $\frac{3}{2}t = w_3 + 2w_2$, so $t_R = \frac{4w_2 + 2w_3}{3}$. Consistency: $t_R \geq 4w_2 \iff w_3 \geq 4w_2$ and $t_R \leq w_3 \iff 4w_2 \leq w_3$, both holding, so S_3 is indeed still straight at impact.

Mode 2, $\frac{w_2}{4} \leq w_3 \leq 4w_2$ (both curve inside the fan): $a_3 + 3t - 2\sqrt{w_2 t} = a_3 + 2\sqrt{w_3 t}$ gives $3\sqrt{t} = 2(\sqrt{w_2} + \sqrt{w_3})$, so $t_R = \frac{4(\sqrt{w_2} + \sqrt{w_3})^2}{9}$. The common fan value at impact is

$$v^* = 3 - 2\sqrt{w_2/t_R} = \frac{3\sqrt{w_3}}{\sqrt{w_2} + \sqrt{w_3}} = 2\sqrt{w_3/t_R},$$

and $1 \leq v^* \leq 2 \iff \frac{w_2}{4} \leq w_3 \leq 4w_2$: the fan is consumed exactly at impact, leaving a clean 3|0 shock.

Mode 3, $w_3 \leq \frac{w_2}{4}$ (S_3 exhausts the fan first, at $4w_3 \leq w_2$): straight 3|1 against straight 1|0: $w_1 + 2t = a_3 + \frac{t}{2} + 2w_3$ gives $\frac{3}{2}t = w_2 + 2w_3$, so $t_R = \frac{2w_2 + 4w_3}{3}$. Both consistency checks hold here too: $t_R \geq 4w_3 \iff w_2 \geq 4w_3$ and $t_R \leq w_2 \iff w_3 \leq \frac{w_2}{4}$, so S_2 really is still straight at impact.

Continuity across $w_3 = 4w_2$ and $w_2 = 4w_3$: the adjacent formulas agree there, giving $4w_2$ at the first boundary and $4w_3$ at the second. In every mode the fan is fully consumed at the instant of impact (modes 1 and 3 destroy it strictly before), so the outcome is always a straight 3|0 shock of speed $\frac{3}{2}$ starting at the impact point, denoted (x^*, t_R) .

Step 5: Compute the second merger time when the right collision comes first ($t_R \leq t_L$)

After S_2 and S_3 merge at the impact point (x^*, t_R) , exactly two fronts remain: S_1 at $\frac{7}{2}t$ (the state between them is identically 3, since the left state of S_2 was always 3) and the 3|0 shock $x^* + \frac{3}{2}(t - t_R)$. They meet when $\frac{7}{2}\tau_2 = x^* + \frac{3}{2}(\tau_2 - t_R)$, that is $2\tau_2 = x^* - \frac{3}{2}t_R$. Now run through the collision modes one at a time.

If the impact was straight 3|2 on straight 2|0 (the regime $w_3 \geq 4w_2$, with $t_R = \frac{4w_2 + 2w_3}{3}$): $x^* = a_4 + t_R$, so $2\tau_2 = w_1 + w_2 + w_3 - \frac{t_R}{2} = w_1 + w_2 + w_3 - \frac{2w_2 + w_3}{3} = w_1 + \frac{w_2 + 2w_3}{3}$.

If both fronts curved inside the fan (the regime $\frac{w_2}{4} \leq w_3 \leq 4w_2$, with $t_R = \frac{4(\sqrt{w_2} + \sqrt{w_3})^2}{9}$): $x^* = a_3 + 2\sqrt{w_3 t_R} = a_3 + \frac{4}{3}\sqrt{w_3}(\sqrt{w_2} + \sqrt{w_3})$ and $\frac{3}{2}t_R = \frac{2}{3}(\sqrt{w_2} + \sqrt{w_3})^2$; with $p = \sqrt{w_2}$ and $q = \sqrt{w_3}$,

$$x^* - \frac{3}{2}t_R = w_1 + p^2 + \frac{4pq + 4q^2}{3} - \frac{2p^2 + 4pq + 2q^2}{3} = w_1 + \frac{p^2 + 2q^2}{3} = w_1 + \frac{w_2 + 2w_3}{3}.$$

Last, for straight 3|1 on straight 1|0 (the regime $w_3 \leq \frac{w_2}{4}$, with $t_R = \frac{2w_2 + 4w_3}{3}$): $x^* = w_1 + 2t_R$, so $2\tau_2 = w_1 + \frac{t_R}{2} = w_1 + \frac{w_2 + 2w_3}{3}$.

All three agree. So, uniformly,

$$\tau_2 = \frac{3w_1 + w_2 + 2w_3}{6} \quad (t_R \leq t_L).$$

Step 6: Compute the second merger time when the left collision comes first ($t_L \leq t_R$)

After S_1 absorbs S_2 , the merged front B has left state 4. Its trajectory depends on where the merger happened.

(a) Merger in the straight phase ($2w_1 \leq 3w_2$): B is a 4|1 shock of speed $\frac{5}{2}$ from the point $(\frac{7w_1}{3}, \frac{2w_1}{3})$, so $x_B(t) = \frac{5}{2}t + \frac{2w_1}{3}$. It reaches the fan's left edge $a_3 + t$ at

$$t_e = \frac{2w_1 + 6w_2}{9} = \frac{2W}{9},$$

and $t_e \geq \frac{2w_1}{3}$ holds exactly under $2w_1 \leq 3w_2$. Inside the fan the trajectory law $x(t) = a_3 + 4t + C\sqrt{t}$ applies (constant left state $c = 4$); matching the entry point gives $t_e = 4t_e + C\sqrt{t_e}$, so $C = -3\sqrt{t_e} = -\sqrt{2W}$:

$$x_B(t) = a_3 + 4t - \sqrt{2Wt}, \quad v_B(t) = 4 - \sqrt{2W/t}.$$

(b) Merger inside the fan ($3w_2 \leq 2w_1 \leq 10w_2$): the 4|fan front through $(\frac{7}{2}t_L, t_L)$ has the same trajectory. Certificate: with $m = \sqrt{2W}$ and $p = \sqrt{w_2}$, one has $a_3 = w_1 + w_2 = \frac{m^2 - 4p^2}{2}$ and $\sqrt{t_L} = m - 2p$, so

$$a_3 + 4t_L - \sqrt{2Wt_L} - \frac{7}{2}t_L = \frac{m^2 - 4p^2}{2} + \frac{(m - 2p)^2}{2} - m(m - 2p) = 0,$$

that is, the curve $a_3 + 4t - \sqrt{2Wt}$ passes through the merger point, and uniqueness of solutions of the in-fan ODE $\dot{x} = \frac{1}{2} \left(4 + \frac{x-a_3}{t}\right)$ does the rest.

(c) Merger in the post-fan phase ($w_1 \geq 5w_2$; here S_2 already consumed the entire fan at $t = 4w_2$): B is a straight 4|2 shock of speed 3 from $(\frac{7}{2}(w_1 - w_2), w_1 - w_2)$: $x_B(t) = 3t + \frac{w_1 - w_2}{2}$.

Now B must absorb S_3 . Three regimes, partitioned by $W = w_1 + 3w_2$ against w_3 .

Regime L-A, $W \leq 2w_3$ (impact on the straight 2|0 shock S_3). In cases (a)/(b), $v_B = 2$ at $t_X = \frac{2W}{4} = \frac{W}{2}$, at $x = a_3 + 2t_X$ (the fan's right edge), and $t_X \leq w_3$ (the regime inequality) guarantees the constant-2 band still separates B from S_3 ; thereafter $x_B = a_3 + 3t - t_X$. Impact with $a_4 + t$: $3t - t_X = w_3 + t$, so $\tau_2 = \frac{w_3 + t_X}{2} = \frac{W + 2w_3}{4}$. In case (c) the same final configuration (4|2 at speed 3 against 2|0 at speed 1) yields $3t + \frac{w_1 - w_2}{2} = a_4 + t$, so $\tau_2 = \frac{w_1 + 3w_2 + 2w_3}{4} = \frac{W + 2w_3}{4}$, the same formula. And the regime inequality $W \leq 2w_3$ holds automatically in case (c): when $w_1 \geq 5w_2$ and the left collision is first, the right-collision modes with $w_3 \leq 4w_2$ are impossible (for $w_3 \leq \frac{w_2}{4}$ one has $t_R \leq w_2 \leq 4w_2 \leq w_1 - w_2 = t_L$, and for $\frac{w_2}{4} \leq w_3 \leq 4w_2$ one has $t_R \leq \frac{4}{9}(3\sqrt{w_2})^2 = 4w_2 \leq t_L$), so necessarily $w_3 \geq 4w_2$ and $t_L \leq t_R$ reads $3(w_1 - w_2) \leq 4w_2 + 2w_3$, whence $2w_3 \geq 3w_1 - 7w_2 = (w_1 + 3w_2) + (2w_1 - 10w_2) \geq W$ using $w_1 \geq 5w_2$.

Regime L-B, $2w_3 \leq W \leq 18w_3$ (impact inside the fan). Both fronts curve: $a_3 + 4t - \sqrt{2Wt} = a_3 + 2\sqrt{w_3t}$ gives $4\sqrt{t} = \sqrt{2W} + 2\sqrt{w_3}$, so

$$\tau_2 = \frac{(\sqrt{2W} + 2\sqrt{w_3})^2}{16}.$$

The common fan value at impact is $v^{**} = 4 - \frac{4\sqrt{2W}}{\sqrt{2W} + 2\sqrt{w_3}} = \frac{8\sqrt{w_3}}{\sqrt{2W} + 2\sqrt{w_3}}$, and $1 \leq v^{**} \leq 2 \iff 2w_3 \leq W \leq 18w_3$: these are exactly the regime boundaries, and again the fan dies exactly at the impact.

Regime L-C, $W \geq 18w_3$ (impact on the 1|0 remnant). Here $4w_3 \leq \frac{2W}{9} = t_e$, so S_3 consumes the whole fan before B arrives and B never enters the fan. This regime forces case (a). Certificate: suppose case (b) or (c), so $2w_1 \geq 3w_2$, together with $t_L \leq t_R$ and $W \geq 18w_3$. Case (c) is excluded at once, since there $W \leq 2w_3$ (the bound established in regime L-A), contradicting $W \geq 18w_3$. In case (b), $w_1 \leq 5w_2$ gives $W \leq 8w_2$, so $w_3 \leq \frac{W}{18} \leq \frac{4w_2}{9}$. If $w_3 \leq \frac{w_2}{4}$, then $t_R = \frac{2w_2 + 4w_3}{3} \leq w_2 \leq t_L$ (the latter since $2w_1 \geq 3w_2$ places $t_L \geq w_2$), contradicting $t_L \leq t_R$. If $w_3 > \frac{w_2}{4}$, then both fronts use their middle branches and $t_L \leq t_R$ means $\sqrt{2W} - 2\sqrt{w_2} \leq \frac{2}{3}(\sqrt{w_2} + \sqrt{w_3})$, which squares to $9w_1 \leq 5w_2 + 2w_3 + 16\sqrt{w_2w_3}$; substituting $w_1 \geq 18w_3 - 3w_2$ (from $W \geq 18w_3$) gives $160w_3 - 32w_2 \leq 16\sqrt{w_2w_3}$, that is $10s^2 - s - 2 \leq 0$ for $s = \sqrt{w_3/w_2}$, forcing $s \leq \frac{1}{2}$ and contradicting $s > \frac{1}{2}$. So case (a) holds, and the 4|1 line meets the 1|0 line: $\frac{5}{2}t + \frac{2w_1}{3} = a_3 + \frac{t}{2} + 2w_3$ gives $2t = \frac{w_1}{3} + w_2 + 2w_3$, so

$$\tau_2 = \frac{w_1 + 3w_2 + 6w_3}{6} = \frac{W + 6w_3}{6},$$

with consistency $\tau_2 \geq 4w_3 \iff W \geq 18w_3$.

Continuity across regime boundaries: at $W = 2w_3$ both L-A and L-B give w_3 ; at $W = 18w_3$ both L-B and L-C give $4w_3$.

Step 7: Resolve $\min(t_L, t_R)$ into explicit width inequalities

Write L_1, L_2, L_3 for the branches of t_L (straight, in-fan, post-fan, with ranges $2w_1 \leq 3w_2$, $3w_2 \leq 2w_1 \leq 10w_2$, $w_1 \geq 5w_2$) and R_1, R_2, R_3 for the modes of t_R (with ranges $w_3 \geq 4w_2$, $\frac{w_2}{4} \leq w_3 \leq 4w_2$, $w_3 \leq \frac{w_2}{4}$). The nine cells compare as follows.

In (L_1, R_1) : $t_R \geq \frac{4w_2+8w_2}{3} = 4w_2$ while $t_L = \frac{2w_1}{3} \leq w_2 < 4w_2$, so L is always first. For (L_1, R_2) , $t_R = \frac{4}{9}(\sqrt{w_2} + \sqrt{w_3})^2 \geq \frac{4}{9}(\sqrt{w_2} + \frac{\sqrt{w_2}}{2})^2 = w_2 \geq \frac{2w_1}{3} = t_L$, so L is always first, with equality only at the corner $w_1 = \frac{3}{2}w_2$, $w_3 = \frac{w_2}{4}$. The cell (L_1, R_3) is the first genuine race: $t_L \leq t_R \iff \frac{2w_1}{3} \leq \frac{2w_2+4w_3}{3} \iff w_1 \leq w_2 + 2w_3$, and here $w_2 + 2w_3 \leq \frac{3}{2}w_2$, so both orders occur and the comparison is w_1 against $w_2 + 2w_3$. (L_2, R_1) is quick: $t_L \leq 4w_2 \leq t_R$, so L is always first. The cell (L_2, R_2) takes more work: $t_L \leq t_R \iff \sqrt{2W} - 2\sqrt{w_2} \leq \frac{2}{3}(\sqrt{w_2} + \sqrt{w_3}) \iff 3\sqrt{2W} \leq 8\sqrt{w_2} + 2\sqrt{w_3} \iff$

$18W \leq 64w_2 + 32\sqrt{w_2w_3} + 4w_3 \iff 9w_1 \leq 5w_2 + 2w_3 + 16\sqrt{w_2w_3}$ (using $18W = 18w_1 + 54w_2$), so the comparison is $9w_1$ against $5w_2 + 2w_3 + 16\sqrt{w_2w_3}$. In (L_2, R_3) , $t_R \leq \frac{2w_2+w_2}{3} = w_2 \leq t_L$, so R is always first. For (L_3, R_1) , $t_L \leq t_R \iff 3(w_1 - w_2) \leq 4w_2 + 2w_3 \iff 3w_1 \leq 7w_2 + 2w_3$, so the comparison is $3w_1$ against $7w_2 + 2w_3$. In (L_3, R_2) : $t_R \leq \frac{4}{9}(3\sqrt{w_2})^2 = 4w_2 \leq w_1 - w_2 = t_L$, so R is always first. And (L_3, R_3) : $t_R \leq w_2 \leq 4w_2 \leq t_L$, so R is always first.

In cells where one branch's validity premise fails (for instance L_3 presumes the fan survives to $4w_2$), the bounds just listed show t_R comes earlier than any possible catch by S_1 , so the conclusion that R is first stands on actual trajectories. Collecting the R -first set and absorbing the always- R cells into the three parametric inequalities: in $\{w_3 \leq \frac{w_2}{4}\}$, the cells (L_2, R_3) and (L_3, R_3) have $w_1 \geq \frac{3}{2}w_2 \geq w_2 + 2w_3$, so the single inequality $w_1 \geq w_2 + 2w_3$ captures all R -first points there. In $\{\frac{w_2}{4} \leq w_3 \leq 4w_2\}$, the cell (L_3, R_2) has $9w_1 \geq 45w_2 \geq 5w_2 + 2w_3 + 16\sqrt{w_2w_3}$ (the right side is maximized at $w_3 = 4w_2$, with value $45w_2$), so $9w_1 \geq 5w_2 + 2w_3 + 16\sqrt{w_2w_3}$ captures all R -first points, while in the cell (L_1, R_2) this inequality never holds strictly ($9w_1 \leq \frac{27}{2}w_2 \leq 5w_2 + 2w_3 + 16\sqrt{w_2w_3}$), consistent with L always first there. In $\{w_3 \geq 4w_2\}$, the inequality $3w_1 \geq 7w_2 + 2w_3 \geq 15w_2$ already forces $w_1 \geq 5w_2$, the cell L_3 , so the single inequality $3w_1 \geq 7w_2 + 2w_3$ is exact. The upshot:

$$t_R \leq t_L \iff \{w_3 \leq \frac{w_2}{4}, w_1 \geq w_2 + 2w_3\} \cup \{\frac{w_2}{4} \leq w_3 \leq 4w_2, 9w_1 \geq 5w_2 + 2w_3 + 16\sqrt{w_2w_3}\} \cup \{w_3 \geq 4w_2, 3w_1 \geq 7w_2 + 2w_3\}$$

The three comparison inequalities are tight exactly on the triple-collision surface: there S_1 reaches S_2 at the very moment S_2 reaches S_3 , and both collisions happen at the position of S_2 , so all three shocks meet at one space-time point, n drops from 3 to 1, and $\mathcal{T} = \{\tau_1, \tau_1\}$ with multiplicity 2. All τ_2 formulas extend continuously to this surface with $\tau_2 = \tau_1$. Try $(w_1, w_2, w_3) = (\frac{23}{9}, 1, 1)$: $t_L = (\frac{10}{3} - 2)^2 = \frac{16}{9}$, $t_R = \frac{16}{9}$, the in-fan impact formula gives $\frac{(\frac{10}{3}+2)^2}{16} = \frac{16}{9}$, and the right-first formula gives $\frac{3 \cdot \frac{23}{9} + 1 + 2}{6} = \frac{16}{9}$.

Step 8: Verify every second merger formula against the conservation line

For large $R > 0$, with flux $f(u) = \frac{u^2}{2}$, $\frac{d}{dt} \int_{-R}^R u dx = f(u(-R, t)) - f(u(R, t)) = \frac{4^2}{2} - 0 = 8$, and $\int_{-R}^R u(x, 0) dx = 4R + 3w_1 + w_2 + 2w_3$. After the last merger the profile is 4 to the left and 0 to the right of the single shock at $x_s(t)$, so the integral equals $4x_s + 4R$, forcing

$$x_s(t) = 2t + \frac{3w_1 + w_2 + 2w_3}{4}.$$

Each of the four second-merger formulas must place its merger point on this line, and direct substitution confirms each one. The right-first point $(\frac{7}{2}\tau_2, \tau_2)$ with $\tau_2 = \frac{3w_1+w_2+2w_3}{6}$: $2\tau_2 + \frac{6\tau_2}{4} = \frac{7}{2}\tau_2$. The straight-impact point $(a_3 + 3\tau_2 - \frac{W}{2}, \tau_2)$ with $\tau_2 = \frac{W+2w_3}{4}$: expanding, $w_1 + w_2 + 3\tau_2 - \frac{W}{2} - 2\tau_2 - \frac{3w_1+w_2+2w_3}{4} = \tau_2 - \frac{W+2w_3}{4} = 0$. The in-fan impact point $(a_3 + 2\sqrt{w_3\tau_2}, \tau_2)$ with $\sqrt{\tau_2} = \frac{\sqrt{2W+2\sqrt{w_3}}}{4}$: both sides equal $w_1 + w_2 + w_3 + \frac{\sqrt{2Ww_3}}{2}$. The remnant-impact point with $\tau_2 = \frac{W+6w_3}{6}$: both sides equal $\frac{13w_1}{12} + \frac{5w_2}{4} + \frac{5w_3}{2}$. Cheap algebra, but it is a genuinely independent net, and all four τ_2 formulas pass it.

Step 9: Check degenerate width limits and assemble the final multiset

As $w_2 \rightarrow 0^+$ the datum becomes the three-level cascade 4, 3, 2, 0 and $t_L \rightarrow w_1$ (post-fan branch), matching the direct computation $\frac{7}{2}t = a_3 + \frac{5}{2}t$ for the 4|3 shock catching the 3|2 shock. As $w_3 \rightarrow 0^+$ the datum becomes 4, 3, 1, 0 and $t_R \rightarrow \frac{2w_2}{3}$ (the $w_3 \leq \frac{w_2}{4}$ mode), matching the 3|1 shock of speed 2 catching the 1|0 shock of speed $\frac{1}{2}$ across the gap w_2 : $\frac{3}{2}t = w_2$.

Collecting everything, the first merger time is $\tau_1 = \min(t_L, t_R)$ with

$$t_L = \begin{cases} \frac{2w_1}{3}, & 2w_1 \leq 3w_2, \\ (\sqrt{2w_1 + 6w_2} - 2\sqrt{w_2})^2, & 3w_2 \leq 2w_1 \leq 10w_2, \\ w_1 - w_2, & w_1 \geq 5w_2, \end{cases}$$

and

$$t_R = \begin{cases} \frac{4w_2 + 2w_3}{3}, & w_3 \geq 4w_2, \\ \frac{4(\sqrt{w_2 + \sqrt{w_3}})^2}{9}, & \frac{w_2}{4} \leq w_3 \leq 4w_2, \\ \frac{2w_2 + 4w_3}{3}, & w_3 \leq \frac{w_2}{4}, \end{cases}$$

where $t_R \leq t_L$ holds exactly on the region $\{w_3 \leq \frac{w_2}{4}, w_1 \geq w_2 + 2w_3\} \cup \{\frac{w_2}{4} \leq w_3 \leq 4w_2, 9w_1 \geq 5w_2 + 2w_3 + 16\sqrt{w_2 w_3}\}$. The second merger time is $\frac{w_1 + 3w_2 + 2w_3}{4}$, $\frac{(\sqrt{2w_1 + 6w_2} + 2\sqrt{w_3})^2}{16}$, or $\frac{w_1 + 3w_2 + 6w_3}{6}$ when the left collision is first (according to $w_1 + 3w_2 \leq 2w_3$, $2w_3 \leq w_1 + 3w_2 \leq 18w_3$, or $w_1 + 3w_2 \geq 18w_3$), and $\frac{3w_1 + w_2 + 2w_3}{6}$ when the right collision is first. On every boundary surface the adjacent formulas agree, so $\mathcal{T}$ is well defined and continuous on all of $(0, \infty)^3$; on the surface $t_L = t_R$ the collision is triple, $\tau_1 = \tau_2$, and the multiset is $\{\tau_1, \tau_1\}$ with multiplicity 2, consistent with $n(\tau_1^-) - n(\tau_1) = 2$.

Final Answer: $\mathcal{T} = \{\tau_1, \tau_2\}, \quad \tau_1 = \min(t_L, t_R), \quad \tau_2 = \begin{cases} \frac{w_1 + 3w_2 + 2w_3}{4}, & t_L \leq t_R \text{ and } w_1 + 3w_2 \leq 2w_3, \\ \frac{(\sqrt{2w_1 + 6w_2} + 2\sqrt{w_3})^2}{16}, & t_L \leq t_R \text{ and } 2w_3 \leq w_1 + 3w_2 \leq 18w_3, \\ \frac{w_1 + 3w_2 + 6w_3}{6}, & t_L \leq t_R \text{ and } w_1 + 3w_2 \geq 18w_3, \\ \frac{3w_1 + w_2 + 2w_3}{6}, & t_R \leq t_L \end{cases}$

Answer

$$\mathcal{T} = \{\tau_1, \tau_2\}, \quad \tau_1 = \min(t_L, t_R), \quad \tau_2 = \begin{cases} \frac{w_1 + 3w_2 + 2w_3}{4}, & t_L \leq t_R \text{ and } w_1 + 3w_2 \leq 2w_3, \\ \frac{(\sqrt{2w_1 + 6w_2} + 2\sqrt{w_3})^2}{16}, & t_L \leq t_R \text{ and } 2w_3 \leq w_1 + 3w_2 \leq 18w_3, \\ \frac{w_1 + 3w_2 + 6w_3}{6}, & t_L \leq t_R \text{ and } w_1 + 3w_2 \geq 18w_3, \\ \frac{3w_1 + w_2 + 2w_3}{6}, & t_R \leq t_L \end{cases}$$

Appendix LaTeX Formatting

Appendix: LaTeX Formatting

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[LaTeX Formatting](#)

[Polynomial Formatting](#)

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This document outlines acceptable LaTeX formatting requirements:

General Math Formatting Guidelines

Use LaTeX notation for mathematical symbols in prompts, final answers, and step rewrites. Avoid using LaTeX for non-mathematical formatting (i.e., bullet points, bold, etc.).

LaTeX Formatting

- $\$...\$$ and $\$...\$ \$$ are allowed
- $\backslash (... \backslash)$ and $\backslash [... \backslash]$ are NOT allowed

Braces

- Avoid use of unnecessary braces that don't render (e.g., $\{\frac{3}{4}\}\cdot\{\frac{5}{8}\}$)
- Make sure braces are escaped when they should be visible (e.g., $\ln\{1,2,3\}$)

Polynomial Formatting

- Always arrange polynomial terms in descending order of exponents

Fractional Expressions

- Use inline fractions (e.g., $\frac{3}{4}$).
- Simplify fractions whenever possible unless the unsimplified form adds context.

Units and Measurements

- Include units in annotations whenever relevant (e.g., 9.8 m/s^2).

Special Characters and Symbols

- Use proper mathematical symbols (e.g., $\leq$ instead of $<=$).

- Define all special symbols used.

Other Style

- No use of `\item[]` for lists
- No use of `**... **` to bold Steps or Titles
- No LaTeX commands only valid in KaTeX (e.g. `\lang` instead of `\langle`)

Rounding and Approximations

If decimal approximations are required in the final answer, rounding instructions must be specified within the prompt (e.g., round to at least four decimal places). The model's rounding choices cannot be considered errors without these instructions.

Uniformity

Maintain consistent formatting for all submissions.

Examples of Accepted LaTeX Formatting

Category	Numerical	LaTeX Equivalent	Rendered Form
Fractions	$1/2$	<code>\frac{1}{2}</code>	$\frac{1}{2}$
	$1 / (2+3)$	<code>\frac{1}{2+3}</code>	$\frac{1}{2+3}$
Infinity	∞	<code>\infty</code>	∞
Exponents	2^3	<code>2^{3}</code>	2^3
	e^2	<code>e^{2}</code>	e^2
	$x^{(1/2)}$	<code>x^{\frac{1}{2}}</code>	$x^{\frac{1}{2}}$

Category	Numerical	LaTeX Equivalent	Rendered Form
	<code>sqrt(2)</code>	<code>\sqrt{2}</code>	$\sqrt{2}$
	<code>sqrt[3]{8}</code>	<code>\sqrt[3]{8}</code>	$\sqrt[3]{8}$
Mathematical Constants	<code>pi</code>	<code>\pi</code>	π
	<code>e</code>	<code>e</code>	e
	<code>i</code>	<code>i</code>	i
Golden Ratio	<code>phi</code>	<code>\phi</code>	ϕ
Trigonometry	<code>sin(pi/2)</code>	<code>\sin(\pi/2)</code>	$\sin(\pi/2)$
	<code>cos(0)</code>	<code>\cos(0)</code>	$\cos(0)$
	<code>tan(pi/4)</code>	<code>\tan(\pi/4)</code>	$\tan(\pi/4)$
Logarithms	<code>ln(2)</code>	<code>\ln(2)</code>	$\ln(2)$
	<code>log10(100)</code>	<code>\log_{10}(100)</code>	$\log_{10}(100)$
	<code>log2(x)</code>	<code>\log_2(x)</code>	$\log_2(x)$
Set Notation	<code>{1, 2, 3}</code>	<code>\{1, 2, 3\}</code>	$\{1, 2, 3\}$
	<code>{}</code>	<code>\emptyset</code>	$\emptyset$
	<code>R=(-∞, ∞)</code>	<code>\mathbb{R}</code>	$\mathbb{R}$
	<code>p, q ∈ Z (integers)</code>	<code>\mathbb{Q}</code>	$\mathbb{Q}$
	<code>N={1, 2, 3, 4, ...}</code>	<code>\mathbb{N}</code>	$\mathbb{N}$

Category	Numerical	LaTeX Equivalent	Rendered Form
	$\{x x>0\}$	<code>\{ x \mid x > 0 \}</code>	$\{ x \mid x > 0 \}$
	$x \in A$	<code>x \in A</code>	$x \in A$
	$x \notin A$	<code>x \notin A</code>	$x \notin A$
	$A \cup B$	<code>A \cup B</code>	$A \cup B$
	$A \cap B$	<code>A \cap B</code>	$A \cap B$
	$A \setminus B$	<code>A \setminus B</code>	$A \setminus B$
	$P(A)$	<code>\mathcal{P}(A)</code>	$\mathcal{P}(A)$
	$\forall x \in A$	<code>\forall x \in A</code>	$\forall x \in A$
	$\exists x \in A$	<code>\exists x \in A</code>	$\exists x \in A$
Combinatorics	$C(5, 2)$	<code>\binom{5}{2}</code>	$\binom{5}{2}$
	$P(5, 2)$	<code>_5P_2</code>	$_5P_2$
	$5!$	<code>5!</code>	$5!$
Integrals	$\int_0^1 f(x) dx$	<code>\int_{0}^{1} f(x) \, dx</code>	$\int_0^1 f(x) dx$
	$\int f(x) dx$	<code>\int f(x) \, dx</code>	$\int f(x) dx$
	$\iint_R f(x, y) dx dy$	<code>\iint_{R} f(x, y) \, dx \, dy</code>	$\iint_R f(x, y) dx dy$
	$\iiint f(x, y, z) dx dy dz$	<code>\iiint f(x, y, z) \, dx \, dy \, dz</code>	$\iiint f(x, y, z) dx dy dz$
	$\oint_C f(z) dz$	<code>\oint_{C} f(z) \, dz</code>	$\oint_C f(z) dz$

Category	Numerical	LaTeX Equivalent	Rendered Form
Matrices	$\begin{bmatrix} a, & b, \\ c, & d \end{bmatrix}$	<code>\begin{bmatrix} a & b \\ c & d \end{bmatrix}</code>	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$
	$\begin{bmatrix} a, & b, & c, \\ d, & e, & f, \\ g, & h, & i \end{bmatrix}$	<code>\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}</code>	$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$
	I	<code>\mathbf{I}</code>	$\mathbf{I}$
	$\det(A)$	<code>\det(A)</code>	$\det(A)$
	A^T	<code>A^{T}</code>	A^T
Closed-Form Equations	$(-b \pm \sqrt{b^2 - 4ac})/(2a)$	<code>\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}</code>	$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
	$(x + y)^n = \sum_{k=0}^n C(n, k) x^{n-k} y^k$	<code>(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k</code>	$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$
	$P(t) = P_0 e^{rt}$	<code>P(t) = P_0 e^{rt}</code>	$P(t) = P_0 e^{rt}$
	$S = a/(1 - r)$	<code>S = \frac{a}{1 - r}</code>	$S = \frac{a}{1 - r}$
Approximations (to be specified in prompts if used)	$\sqrt{2} \approx 1.414$	<code>\sqrt{2} \approx 1.414</code>	$\sqrt{2} \approx 1.414$
	$\pi \approx 3.1416$	<code>\pi \approx 3.1416</code>	$\pi \approx 3.1416$
	$\ln(2) \approx 0.693$	<code>\ln(2) \approx 0.693</code>	$\ln(2) \approx 0.693$
Vectors	$[1, 2, 3]$	<code>\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}</code>	$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$

Category	Numerical	LaTeX Equivalent	Rendered Form
	$[1 \ 2 \ 3]$	<code>\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}</code>	$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$
Coordinates	$(3, \ 4)$	<code>\[(3, \ 4) \]</code>	$(3, 4)$
	$(x, \ y, \ z)$	<code>\[(x, \ y, \ z) \]</code>	(x, y, z)
Intervals	$(\emptyset, \ 1)$	<code>\[(\emptyset, \ 1) \]</code>	$(\emptyset, 1)$
	$[0, 1]$	<code>\[[0, \ 1] \]</code>	$[0, 1]$
	$[0, 1)$	<code>[0, \ 1)</code>	$[0, 1)$

Code of Conduct

Expert Code of Conduct

 **PLEASE READ:** It is expected that **all** experts adhere to Project Rainier's Code of Conduct. If an expert is found to be within violation of **any** of the following reasons, this could result in immediate dismissal from the project and/or any future collaboration with Snorkel:

Spam:

Your submissions reflect a pattern of short, repetitive, or templated responses that don't meet our standards for original and thoughtful engagement. High-volume, low-value content that disrupts peer workflows and degrades overall task quality.

Plagiarism:

Some of your submissions closely resemble external content or other experts' work. This includes copied prompts/responses, published materials, or reworded internal content. All work must be original and authentic.

LLM Usage:

*Any clear indication of use of a language model in a way that is not aligned with this project's guidelines (i.e., **outside of** using external spell-checkers and grammar-checking tools). LLM-generated prompts, responses, or feedback given in a task.*

Repeated Rejections:

*Work that has **repeatedly** failed peer review due to quality issues, despite being provided support and corrective actions by the Rainier Team. While occasional feedback is normal, a consistent pattern of quality issues affects downstream performance.*

Low AHT:

Completion times which suggest that you are prioritizing speed over quality. We encourage you to take your time and be thorough. We expect careful, detailed work that meets our standards.

Frustrated Expert:

Messages to the Snorkel Team or fellow ECs that signal increasing frustration or negativity, via Slack or on the Expert Platform. To preserve group cohesion and safety, we expect ECs to maintain respectful and professional relations with Snorkel staff and other ECs at all times.

Insidious Expert:

Interactions that suggest attempts to recruit, solicit, or encourage users to engage with external platforms or agendas. This behavior violates our community guidelines and must cease immediately.

Inappropriate Expert:

Communication or behavior that is seen as being inappropriate, disrespectful, or unprofessional. This includes profane and obscene language, or threats against Snorkel staff and/or other ECs.